

What Should a Self-Teacher See? Privileged Context Design for On-Policy Self-Distillation

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More privileged information does not always make a better teacher. We study this tension in on-policy self-distillation (OPSD), where a self-teacher scores the student’s own rollouts under privileged context, conventionally a complete reference solution that bundles the final answer with one particular reasoning path. Holding the student view and training fixed within each scale, we compare that default against three abstractions compiled offline, a *named strategy*, a *method-independent framing*, and a *problem category*, and against an *answer-only* control that keeps the destination but removes the path. In the primary runs on competition mathematics, the best intermediate contexts improve the in-domain peak mean over the full solution by 1.4 points at 4B and 1.6 at 8B, while storing an order of magnitude fewer hint tokens. Comparisons across three seeds also show positive mean gains for the framing and category contexts at both scales. Answer-only conditioning remains competitive in the primary runs, within 0.2 points of the full solution at these scales. The preferred context varies with student scale and task. Initial teacher–student KL does not order downstream performance. What a self-teacher should see is therefore not everything it could, but the level of abstraction its student can still act on.

1. Introduction

On-policy self-distillation (OPSD) promises dense reasoning supervision without a more capable online teacher. The student samples a trajectory from the problem alone, and a frozen copy of its initial checkpoint provides next-token distributions at the same prefixes while observing teacher-only information, such as a verified solution [1]. Whereas conventional on-policy distillation typically derives its teacher advantage from greater model capability [2, 3], OPSD relies on *information asymmetry* instead, which makes the design of this privileged context part of the supervision mechanism.

More detailed context, however, need not provide more useful supervision: under peak reporting on competition mathematics, replacing the full solution with its final answer alone changes 4B and 8B aggregate scores by less than 0.2 points. A full worked solution supplies an explicit procedure but may anchor the teacher’s guidance to a particular derivation, whereas an abstract hint or bare answer offers less procedural support and more latitude in how the problem is solved. These choices can change which continuations the teacher favors, not merely how strongly its distribution differs from the student’s. This raises a central question: *at what level of abstraction should a self-teacher see a reference solution, and how should that choice depend on the student and task?*

Prior work demonstrates the value of privileged solutions for self-distillation [1], while also identifying risks of reduced exploration and reference-path bias [4, 5]. ATESD varies the visible reference prefix while retaining the answer [6], and a broader reference-type comparison includes abstract hints [7]. Using a training-free gradient-alignment proxy, Armandpour et al. [8] report no universally best distillation context and rank summarized demonstrations above raw ones for their larger student. What remains unclear is how adjacent semantic abstractions without explicit solution steps affect student performance across scales and tasks.

We study this question through controlled interventions on the context supplied to a fixed self-teacher

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* Code is available at <https://github.com/tiankanghui/Self-Teacher-Context-Design>.

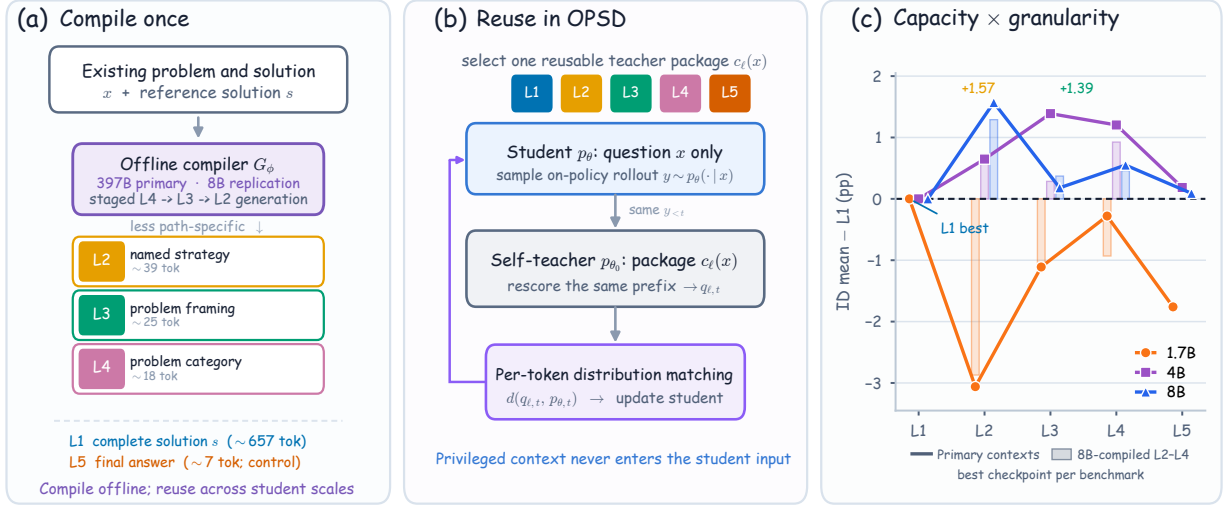


Figure 1: Study overview. (a) An offline compiler produces reusable contexts from coarse to fine (L4→L3→L2); L1 is the original solution and L5 its final answer. Labels give mean token counts. (b) The student samples y from x alone, while a frozen copy of the base model scores the same prefixes with additional context $c_\ell(x)$; only the student is updated. (c) Peak aggregate scores relative to L1 for the primary compiler (solid lines, seed 42) and the single-run Qwen3-8B replication (pale bars). The two compilers agree at the family level but not on the best intermediate level. Replications of L1, L3, and L4 across three seeds appear in Table 3.

(Figure 1). Starting from the same problem–solution pairs, an offline compiler constructs three progressively more abstract context packages: a named strategy (L2), a method-independent framing (L3), and a problem category (L4), all generated under contracts that prohibit the final answer. We compare these with the full solution (L1), an answer-only condition (L5), and a control that shortens the reference to a fixed 10% prefix plus the answer without semantic rewriting. Across three Qwen3 student scales [9], we hold the student prompt, frozen teacher checkpoint, training set, rollout procedure, and loss fixed within each scale, and we evaluate on three competition mathematics benchmarks and four transfer benchmarks. We assess context utility through student performance after training and use initial teacher–student KL as a separate diagnostic.

Our contributions are:

- **Concise semantic contexts can outperform full solutions.** In the primary runs, the best intermediate context exceeds the full solution by 1.4 and 1.6 points in the 4B and 8B in-domain peak means while storing an order of magnitude fewer hint tokens (Table 1). Across three seeds, L3 and L4 show positive mean gains over L1 at 4B and 8B under all four reporting rules. Recompiling the contexts with Qwen3-8B also preserves the peak advantage at the larger scales. The preferred representation varies with student scale and task.
- **Answer-only context remains competitive at the larger scales.** Answer-only conditioning comes within 0.2 points of the full solution’s in-domain peak mean at 4B and 8B, though it trails by 1.76 points at 1.7B.
- **Distributional displacement does not order teaching utility.** The full solution induces the largest initial teacher–student KL at every scale, yet does not yield the highest in-domain peak mean at 4B or 8B. Item-level analysis shows context redistributing which problems the trained student solves. Privileged context is therefore best judged by the students it trains.

2. Related Work

On-Policy Distillation. Knowledge distillation transfers model predictions or reasoning traces [10–12], and bootstrapped methods learn from the model’s own successful rationales [13]. On-policy distillation instead supplies token-level teacher feedback on trajectories sampled from the student’s current policy [2, 3, 14, 15], with later work refining the objective [16–18], conditioning the self-teacher on demonstrations [19], or combining it with an external teacher [20]. Our baseline is OPSD, which conditions the frozen self-teacher on

a complete reference solution [1]; that choice of context is the variable we study.

Privileged Information and Reasoning Guidance. Privileged context gives the teacher access to information withheld from the student. This makes OPSD an instance of learning using privileged information: the teacher sees training-time signals the student will not receive at inference [21, 22]. Context distillation internalizes prompt-induced behavior into parameters [23]. The privileged signal takes many forms: feedback and self-generated traces [24, 25], contextual instructions such as concision [26, 27], recoverable visual cues and other forms of privileged information [28–30], and majority-vote pseudo-solutions requiring no external supervision [31]. It also enters at different stages, through partial references during generation [32] or step-level process supervision [33]. Given a solution, designing the teacher context requires deciding which parts to expose and in what form.

Reference Exposure. ATESD learns how much of a reference reasoning prefix to expose while retaining the final answer [6]. Complementary analyses examine the supervision induced by reference conditioning, identifying confidence miscalibration [34], failures under long generation budgets [4], and per-token bias toward the supplied solution [5]. Reference controls distinguish problem-specific information from context-induced teacher behavior [35]. Beyond exposure length, Shrestha and Tessier [7] compare teacher modes and reference types, including a method-level abstract-hint condition, and analyze how teacher supervision aligns with changes in student predictions. Using a training-free gradient-alignment proxy, Armandpour et al. [8] likewise report no universally best context, ranking summarized demonstrations above raw ones for their larger student. PS-OPSD recasts the reference as a problem-solving structure with a selected path of state transitions [36]. We instead compare three adjacent abstraction levels that omit explicit solution steps, training one student per level.

3. Privileged Context as a Controlled Intervention

We treat the context package supplied to the teacher as the intervention and evaluate its utility after student training. This section formalizes the comparison, describes on-policy supervision and staged context construction, and specifies the diagnostic tests and controls.

3.1. Problem formulation

Let $\mathcal{D} = \{(x_i, s_i)\}_{i=1}^N$ be a training set of problems and their reference solutions. A context policy c_ℓ maps each training problem to a precomputed prompt for the teacher:

$$c_\ell(x) = B_\ell(x, h_\ell(x)), \quad (1)$$

where $h_\ell(x)$ is a representation constructed from the problem and its reference solution, and B_ℓ is the prompt template that combines the original problem, the representation, and the wording that introduces it. We refer to this introductory wording as the *bridge*. The student receives x alone; the context policy affects only the supervision supplied by a frozen teacher.

In on-policy supervision, the teacher scores the student’s own prefixes, which may depart from the reference derivation. We therefore test which representation of that reference provides the most useful supervision.

Our target is the utility of a context *after training*, rather than a property of the teacher distribution alone. Let $\theta_{\ell, \xi, k}$ denote the student checkpoint at step $k \in \mathcal{S}$, trained from the shared initialization θ_0 under context c_ℓ and training randomness ξ , with the remaining pipeline fixed; θ_0 also serves as the frozen teacher checkpoint. For a benchmark $\mathcal{B}$ and a specified reporting rule r , we define

$$U_{\mathcal{B}, r}(c_\ell; \theta_0) = \mathbb{E}_\xi[r(\{\text{Score}_\mathcal{B}(\theta_{\ell, \xi, k})\}_{k \in \mathcal{S}})]. \quad (2)$$

Here, r may report a peak score, a fixed endpoint, or another checkpoint aggregate. Dependence on the fixed training set and pipeline is suppressed. Context design then amounts to maximizing $U_{\mathcal{B}, r}$ over a candidate set of context policies. In the primary study, each training run uses a fixed context level. We vary this level across runs to test whether the best representation changes with student scale, evaluation task, or reporting rule, instead of assuming a universally optimal context. Supplementary experiments examine heterogeneous assignment and learned routing (Section 4.2).

3.2. How context enters on-policy self-distillation

Context changes the teacher’s supervision targets without changing its parameters. At each update, the current student samples a trajectory $y = (y_1, \dots, y_T) \sim p_\theta(\cdot | x)$, and a frozen copy of the initial checkpoint scores the same trajectory prefixes with privileged context:

$$p_{\theta,t}(\cdot) = p_\theta(\cdot | x, y_{<t}), \quad q_{\ell,t}(\cdot) = p_{\theta_0}(\cdot | c_\ell(x), y_{<t}). \quad (3)$$

At a fixed problem and prefix, changing $c_\ell(x)$ changes the effective teacher distribution even though θ_0 is unchanged.

Every condition uses the same clipped full-vocabulary objective. Writing d for this shared objective, the loss for each trajectory is

$$\mathcal{L}_\ell(\theta; x, y) = \frac{1}{T} \sum_{t=1}^T d(q_{\ell,t}, p_{\theta,t}). \quad (4)$$

Appendix A.1 gives the exact clipping and reduction. The teacher is frozen and the sampled trajectory is fixed when differentiating this loss, so a context change can alter both the magnitude and the direction of $-\nabla_\theta \mathcal{L}_\ell$. Greater teacher–student disagreement need not imply a more useful update.

The context package is used only for teacher scoring: the student never receives the hint or bridge, during rollout generation or at evaluation. Within each primary comparison we hold the student initialization, student prompt, frozen teacher checkpoint, training set, rollout procedure, training budget, and loss fixed. What is fixed is the *rollout procedure*, not the realized trajectories: as students learn under different contexts, their subsequent on-policy trajectories can diverge, and this feedback is part of the end-to-end effect being evaluated. Following Zhao et al. [1], the teacher scores with Qwen3’s thinking template while student rollouts use the non-thinking template (Appendix A.1), an interface difference common to L1–L5.

3.3. Constructing matched semantic contexts

We compare representations that vary in their intended specificity to the reference reasoning path (Figure 1(a)). L1 provides the complete reference solution. L2 names a method, theorem, or tool without working through solution steps; L3 gives a method-independent framing of the problem; L4 gives only its mathematical domain and object category. All three contracts prohibit the final answer. L5 supplies only the final answer as reference content and serves as the answer-only control.

We define the L1–L4 ordering through the semantic contracts used to generate L2–L4 (Appendix B.1), and audit compliance in the resulting contexts (Table 7). Each of L2–L4 is newly written rather than a truncation of L1, so the ordering expresses intended abstraction rather than nested text, and our comparisons treat each representation as a complete context package.

Staged Semantic Compilation. Let $s(x)$ denote the existing reference solution and G_ϕ an offline compiler. We construct L2–L4 by refining from coarse to fine, each call receiving the problem, the reference solution, and every coarser representation already produced:

$$h_\ell(x) = G_\phi^{(\ell)}(x, s(x), \{h_{\ell'}(x)\}_{\ell' < \ell \leq 4}), \quad \ell = 4, 3, 2, \quad (5)$$

so that h_4 is compiled from the problem and solution alone. Passing the coarser outputs to finer calls makes the intended semantic boundaries explicit. L1 uses the original solution and L5 extracts its final answer, so neither requires compilation; Appendix B.1 shows all conditions for one problem.

The primary contexts are compiled with Qwen3.5-397B-A17B [37], and we repeat the same staged contracts with Qwen3-8B in thinking mode to test sensitivity to the choice of compiler. Both compilers receive the existing reference solutions, and all contexts are precomputed, so neither compiler is queried during OPSD training.

3.4. Utility estimation and diagnostic tests

We evaluate context utility through student scores after training under the reporting rule in Equation 2. For the in-domain comparison, we train students under L1, L3, and L4 with three seeds at each scale and apply

four checkpoint aggregation rules, two of them selection-free. For L2 and L5, the transfer evaluations, and the supplementary controls, we report one training run per condition rather than averages across seeds. Where reported, resampling matched items quantifies evaluation uncertainty, not variability across training runs. Peak and selection-free rules define different estimands and are distinguished throughout.

Distributional Discrepancy. We separately measure how much a context changes the frozen base model’s next-token distribution. Within each scale, let μ_0 denote the shared empirical distribution of problem–prefix pairs from fixed rollouts of the base model. We compute

$$D(c_\ell) = \mathbb{E}_{(x, y_{<t}) \sim \mu_0} [D_{\text{KL}}(q_{\ell, t} \parallel p_{\theta_0, t})], \quad (6)$$

where $p_{\theta_0, t}(\cdot) = p_{\theta_0}(\cdot \mid x, y_{<t})$. Using the same prefixes from the base model across contexts avoids mixing changes induced by context with differences in trained students’ rollout distributions. This diagnostic measures distributional discrepancy; it is not the clipped training objective, an update norm, or a direct measure of teacher quality.

Empirical Predictions. We test three deliberately strong baseline predictions, none of which is assumed by the formulation. *Full-solution dominance* predicts that L1 yields the greatest downstream utility. *Discrepancy-based ordering* predicts that contexts with larger $D(c_\ell)$ also yield greater utility at a fixed student scale, task, and reporting rule. *Context invariance* predicts that one context is preferred across student scales, tasks, and reporting rules; we test it through ranking changes across scales and benchmarks, with item-level gains and losses relative to L1 describing how a given ranking arises.

3.5. Intervention validity and scope

In the primary comparisons, we vary the complete context package while holding the source problem–solution pairs and the training recipe fixed within each scale, so context is the only variable. The packages can still differ in semantic content, length, answer access, and, in the primary runs, bridge wording. Hints generated from the same reference solutions can also differ in quality. These comparisons therefore evaluate the packages as a whole rather than isolating the causal effect of semantic abstraction.

L1 versus L2–L4 tests whether exposing the complete derivation is preferable to the intermediate packages. Comparisons within L2–L4 test the utility of adjacent semantic roles, and L5 tests supervision from the final answer alone.

Across scales the student and training recipe change together, including the effective batch size (Table 4), which makes the scale comparison one between trained systems rather than parameter count alone. Section 4.2 reports controls that each narrow a specific alternative explanation.

Context Length. We distinguish the size of the reusable hint from the length of the teacher’s context prompt. Under the Qwen3 tokenizer, the primary L2–L4 hints average 38.9, 25.2, and 17.7 tokens against 657.3 for L1, a 16.9–37.1 $\times$ reduction in stored hint content; including the problem and bridge, the context prompts average 225.5, 212.7, and 203.3 tokens against 835.0, a 3.7–4.1 $\times$ reduction (Appendix Figure 4(a); Table 6). Both quantify context cost rather than semantic content.

Content Audits. Full-corpus scans find no empty hints, generation-error sentinels, or lexical answer matches for either compiler. A matched audit of 1,200 problems finds no direct answer copying, but does detect answer-equivalent leakage: 1.42% at L2 and 0.33% at L3 for the primary compiler, rising to 4.17% and 2.58% for Qwen3-8B, with L4 at 0.08% and 0.17% (Table 7). Appendix B.3 gives the definitions and prompts for this model-assisted audit.

4. Experiments

We train Qwen3-1.7B, 4B, and 8B [9] for 200 steps with LoRA [38] on the 29,434-problem OPSD mathematics pool [1, 39], checkpointing every 25 steps. At each scale, the student and the frozen self-teacher start from the same checkpoint, and the primary L1–L5 runs share the data order, rollout procedure, objective, and recipe, with no tuning by context level (Appendices A.1 and B.2). Base denotes the unadapted checkpoint.

Table 1: In-domain peak Avg@12 (%), primary compiler for L2–L4. All entries are the primary run (seed 42); Table 3 reports replication of L1, L3, and L4 across three seeds. Mean averages separately selected benchmark peaks; Base is evaluated once. Bold marks the highest score among the trained conditions in each row. The vs. L1 rows give each column’s mean minus the mean for the full solution in percentage points, green above L1 and red below. Selected steps are listed in Appendix C.3.

Model	Benchmark	Base	L1: solution	L2: strategy	L3: framing	L4: category	L5: answer
1.7B	AIME24	49.72	57.22	56.11	56.94	59.72	57.78
	AIME25	36.39	43.33	38.33	41.39	43.06	41.67
	HMMT25	21.67	31.11	28.06	30.00	28.06	26.94
	Mean	35.93	43.89	40.83	42.78	43.61	42.13
	vs. L1	−7.96	ref.	−3.06	−1.11	−0.28	−1.76
4B	AIME24	75.00	76.94	76.94	77.50	76.94	75.28
	AIME25	64.44	68.61	69.44	69.72	69.17	70.56
	HMMT25	41.39	44.72	45.83	47.22	47.78	45.00
	Mean	60.28	63.43	64.07	64.81	64.63	63.61
	vs. L1	−3.15	ref.	+0.65	+1.39	+1.20	+0.19
8B	AIME24	75.28	78.89	79.72	79.44	80.56	78.61
	AIME25	66.94	73.06	74.17	71.39	71.39	72.78
	HMMT25	44.17	48.06	50.83	49.72	49.72	48.89
	Mean	62.13	66.67	68.24	66.85	67.22	66.76
	vs. L1	−4.54	ref.	+1.57	+0.19	+0.56	+0.09

Table 2: No context leads across the whole transfer suite. For each scale and benchmark, *Winner* is the highest-scoring context at step 200 (primary runs, seed 42) and Δ its margin over L1 in percentage points; – marks the two cells where L1 itself wins. The last row counts how many distinct contexts win the four benchmarks at that scale. Full scores are in Appendix Table 9.

Benchmark	1.7B		4B		8B	
	Winner	Δ	Winner	Δ	Winner	Δ
MT-AIME-7Lang	L3	+1.71	L4	+2.10	L2	+1.71
GPQA-Diamond	L3	+0.81	L2	+1.21	L3	+1.41
ZebraLogic	L1	–	L4	+0.40	L3	+1.70
AutoLogi	L4	+0.19	L5	+0.38	L1	–
Distinct winners	3		3		3	

Avg@12 averages correctness over twelve sampled completions per problem. Following the original OPSD convention, we report each in-domain (ID) benchmark’s best Avg@12 across the eight checkpoints. The *ID peak mean* averages these separately selected peaks on AIME24, AIME25, and HMMT25, so it need not represent a single checkpoint; we also report the selection-free step-200 mean and every checkpoint (Appendix C.3). At step 200, transfer evaluation uses MT-AIME2024-7Lang [40] (macro Avg@12 over seven languages), AutoLogi-EN [41] (verifier accuracy), GPQA-Diamond [42] (Avg@10), and ZebraLogic-grid [43] (exact puzzle accuracy; Appendix A.2). All score differences are in percentage points.

4.1. Context utility varies with student scale and task

In-domain Performance. The best intermediate context exceeds the full solution by 1.39 points at 4B and 1.57 at 8B, while L1 keeps the highest peak mean at 1.7B, 0.28 ahead of L4 (Table 1). The winning level does not shift monotonically toward coarser context with scale. At step 200, L1 still leads at 1.7B, while at 4B both L2 and L4 reach 62.59 against 60.93 for L1, and at 8B L3 reaches 65.37 against 65.28 (Appendix C.3).

Transfer Rankings Vary by Task. At step 200, different transfer tasks favor different contexts (Table 2), and this variation appears even within a single scale: at 1.7B, L3 leads on MT-AIME and GPQA while L1 leads

Table 3: In-domain differences from L1, paired by training seed (42–44). Cells show mean $\pm$ sample SD in percentage points; ($n/3$) counts strictly positive differences. SD describes variation across seeds, not a confidence interval. Peak selects each benchmark’s best checkpoint; best common selects one checkpoint per run by the three-benchmark mean. Green/red indicate positive/negative mean differences. Absolute scores are in Appendix C.3.

Student	Contrast	Checkpoint selected by score		Selection-free	
		Per-bench. peak	Best common	All-ckpt. mean	Step 200
1.7B	L3 – L1	-0.12 ± 0.86 (2/3)	-0.12 ± 0.47 (1/3)	-0.14 ± 0.70 (1/3)	-0.83 ± 2.33 (1/3)
	L4 – L1	0.00 ± 0.33 (1/3)	$+0.06 \pm 0.05$ (2/3)	-0.32 ± 0.68 (1/3)	-1.33 ± 2.04 (1/3)
4B	L3 – L1	$+0.71 \pm 0.69$ (2/3)	$+0.90 \pm 0.23$ (3/3)	$+1.03 \pm 0.15$ (3/3)	$+1.67 \pm 0.73$ (3/3)
	L4 – L1	$+0.96 \pm 0.21$ (3/3)	$+1.64 \pm 0.60$ (3/3)	$+1.01 \pm 0.32$ (3/3)	$+1.05 \pm 1.32$ (2/3)
8B	L3 – L1	$+0.25 \pm 0.65$ (2/3)	$+0.62 \pm 0.44$ (3/3)	$+0.49 \pm 0.34$ (3/3)	$+0.28 \pm 0.32$ (3/3)
	L4 – L1	$+0.49 \pm 0.37$ (3/3)	$+0.80 \pm 0.81$ (2/3)	$+0.62 \pm 0.13$ (3/3)	$+0.22 \pm 1.05$ (1/3)

on ZebraLogic. Because all conditions are evaluated at step 200, these ranking changes do not result from selecting different checkpoints for different conditions.

Answer-only Context. L5 is within 0.2 points of L1’s ID peak mean at 4B and 8B, but trails L1 by 1.76 points at 1.7B (Table 1). Answer access without supplied reasoning is therefore competitive without reaching the highest observed utility: the leading intermediate context exceeds L5 at both larger scales, and on ZebraLogic at 8B, L3 scores 88.10 against 85.90.

4.2. Robustness and alternative explanations

Checkpoint Aggregation and Training Seeds. We replicate the L1, L3, and L4 in-domain comparison with three training seeds at every scale and aggregate each run under four checkpoint rules (Table 3). At 4B and 8B, the mean paired gains of both L3 and L4 over L1 are positive under all four rules, and under the selection-free mean across all checkpoints every individual seed favors both intermediate levels. L4 attains a positive paired peak difference in all three seeds at both scales, whereas L3 does not at 4B seed 43 or 8B seed 44; at the endpoint L3 leads in all six runs and L4 in three. At 1.7B, the mean paired peak differences are -0.12 and 0.00 points for L3 and L4, respectively; the two selection-free rules favor L1 on average. The endpoint at step 200 carries the largest paired standard deviation in five of the six contrasts. This is the pattern the remaining controls follow: they reproduce the contrast between full and intermediate contexts more consistently than any particular winner among L2–L4.

Compiler Replication. Regenerating L2–L4 with Qwen3-8B, under the same semantic contracts and training conditions, preserves the ID peak ordering between full and intermediate contexts: L1 leads at 1.7B, and intermediate contexts exceed it at 4B and 8B. At 8B the compiler and student have the same nominal parameter count. The preferred intermediate changes at 4B, from L3 to L4. The 8B compiler also shows higher L2/L3 answer-equivalent leakage, so the packages are not identical in quality; its L4 leakage is only 0.17%, however, and L4 likewise trails L1 at 1.7B and exceeds it at 4B and 8B (Appendix C.2; Table 7).

Semantic Rewriting vs. Prefix Truncation. A control using a 10% reference prefix plus the answer, inspired by Han et al. [6], tests whether shortening the reference without semantic rewriting recovers the observed gains. In one 4B run, the prefix control reaches an ID peak mean of 63.89, above L1 and L5 but below all three intermediate contexts, with L3 and L4 ahead by 0.93 and 0.74 points (Appendix C.2, Table 13). At step 200 it nearly matches L3 (61.67 against 61.76), though L4 remains 0.93 points ahead. Semantic rewriting therefore yields the higher peaks here, and like the other conditions, the prefix differs from L2–L4 in answer access as well as length (Section 3.5).

Bridge Wording and Rollout Length. With a bridge shared across levels, every intermediate context exceeds L1 by 1.0–2.3 ID peak points (Table 20), and doubling the rollout budget from 1,024 to 2,048 tokens leaves every intermediate context at or above L1 on the peak mean (Appendix C.3). Neither bridge wording nor the original length limit is therefore sufficient to explain the contrast.

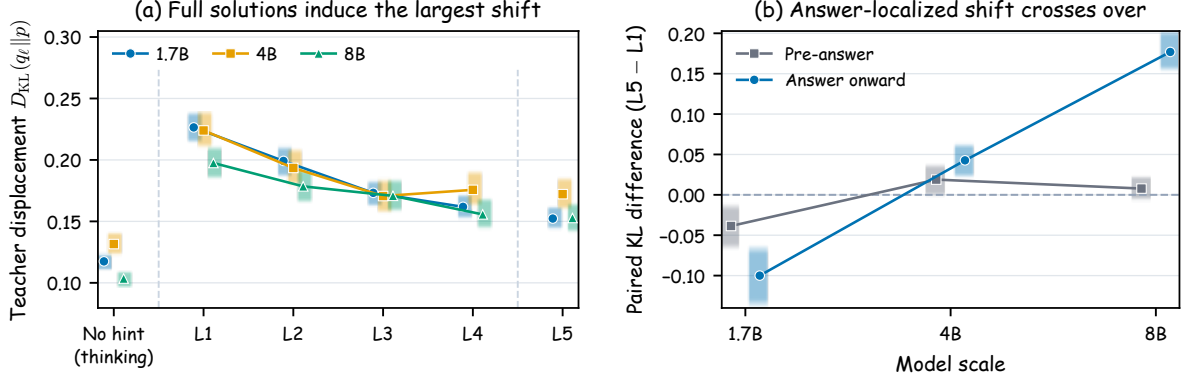


Figure 2: Distributional discrepancy induced by context at initialization, measured on fixed rollouts from the base model with unclipped full-vocabulary KL (95% bootstrap intervals clustered by problem). **(a)** Mean teacher-to-student KL per context; the no-hint interface and L5 are reference conditions outside the L1–L4 hierarchy. **(b)** Paired L5 – L1 differences in pre-answer and answer-onward windows around the final \boxed marker (Appendix D.1).

Mixtures and Routing. Heterogeneous context assignment does not consistently improve on a fixed level. Across balanced mixtures and four routing policies, no run exceeds the best fixed-level ID peak, though some exceed the best fixed-level mean at step 200 (Appendix E).

4.3. Distributional discrepancy does not order context utility

For each scale, the unadapted base model generates two rollouts on each of 600 training problems using the exact student prompt. We rescore identical completion tokens under the student prompt, the L1–L5 teacher packages, and a reference using the teacher interface without a hint, computing unclipped full-vocabulary $D_{KL}(q_{\ell,t} || p_{\theta_{0,t}})$ on shared prefixes as in Equation 6 (Appendix D.1).

L1 has the largest mean KL at every scale (Figure 2(a)). Every package displaces the base distribution well beyond the no-hint interface, so the displacement reflects the supplied context rather than the teacher template alone. Intermediate contexts nevertheless achieve higher 4B and 8B ID peak means with smaller initial discrepancies, so these KL values do not monotonically order the observed peak utilities. Training-free analysis points the same way, reporting only weak within-path correlations between teacher–student divergence and the usefulness of the resulting gradient [8].

Answer-Aligned Differences also Vary with Scale. These means average over all token positions, so we also examine the KL around the final answer. On the subset of rollouts aligned at the final boxed answer, the answer-onward L5 – L1 KL difference is -0.100 at 1.7B but $+0.043$ and $+0.177$ at 4B and 8B (Figure 2(b)). Answer-only context therefore displaces the answer region less than the full solution at the smallest scale and more at the larger ones, as a localized distributional comparison rather than a measure of answer correctness.

4.4. Context changes which problems are solved

On MT-AIME, GPQA, and ZebraLogic, we compare L1 with the highest-scoring intermediate condition at step 200 for each scale–task pair. Let $\hat{a}_i^{L1}$ denote L1’s fraction of correct completions on item i . We classify items as unsolved ($\hat{a}_i^{L1} = 0$), hard ($(0, 0.25]$), medium ($(0.25, 0.75]$), or easy ($(0.75, 1]$); ZebraLogic’s binary outcomes populate only the unsolved and easy buckets. The intermediate condition is selected on the same evaluation used for the comparison, so these analyses describe outcome differences rather than evaluate a selection policy.

Gains are Not Uniform across Difficulty Buckets. Intermediate contexts recover some items unsolved by L1 and improve accuracy in the medium bucket, while sometimes reducing accuracy in L1’s easy bucket (Figure 3; Appendix Table 18). To check dependence on using L1 to define difficulty, we also group items by Base success rates, keeping the same thresholds and the same already selected intermediate condition, so only the

stratification variable changes. Across the six MT-AIME and GPQA scale–task pairs, differences in the medium bucket remain positive at 1.14–10.12 points, whereas differences in the easy bucket range from -0.77 to $+1.65$ points.

Solved-Set Differences Explain the ZebraLogic Contrast.

The fraction of puzzles unsolved by L1 that the selected intermediate rescues rises from 18% at 1.7B to 35% at 8B, while losses on previously solved puzzles shrink. The net accuracy difference changes from -0.70 to $+1.70$ points (Appendix Figure 8). At 1.7B, therefore, recovering some previously unsolved puzzles is compatible with lower overall accuracy; at 8B, the balance of gains and losses favors the intermediate context.

Behavioral Shifts Vary across Scales. We additionally compare model-judged behaviors on matched AIME24 trajectories, selecting the intermediate context by its score on the full AIME24 evaluation at step 200 and including L5 as a separate contrast. Per scale, relative to L1, backtracking rises at 1.7B and falls at 4B and 8B, while explicit correction rises at 1.7B, is unchanged at 4B, and falls at 8B (Table 19); Figure 10 pools these scales into a single aggregate (Appendix D.2). Because these shifts differ across scales, the pooled estimate should not be read as showing behavioral invariance. Neither the per-scale nor the pooled results support an account in which intermediate contexts help by consistently increasing observable backtracking or correction.

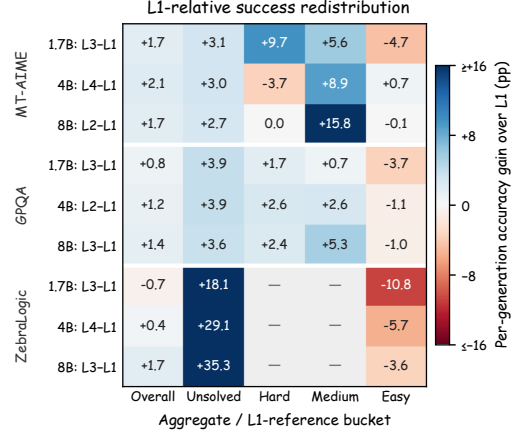


Figure 3: Per-generation accuracy differences at step 200 between L1 and the highest-scoring intermediate context for each scale–task pair, by difficulty bucket under L1. Positive values favor the intermediate context.

4.5. Interpretation and limits

One explanation for these patterns is that full solutions and abstract contexts place different demands on the self-teacher. L1 supplies an executed procedure, which may help when the model cannot reconstruct one from a sparse cue but may also anchor its guidance to the reference derivation, whereas L2–L4 leave more of the procedure to the self-teacher when it evaluates prefixes generated by the student. Teacher and student capacity covary through the shared base checkpoint, so varying the compiler does not separate the teacher’s ability to use the context from the student’s ability to learn from the resulting targets. The analyses at step 200 characterize differences in item-level success and observable reasoning behavior; they do not identify what produces the separately selected peak gains (Section 3.5; Appendix C.4).

5. Conclusion

In OPSD, privileged context serves as a tunable supervision interface, shaping the targets a self-teacher provides on the student’s own trajectories. What, then, should a self-teacher see? Our comparisons show that concise semantic abstractions can outperform full solutions, with the preferred representation depending on student scale and task. A larger initial teacher–student KL does not guarantee better learning outcomes. Item-level analyses further show that context choice produces gains on some problems and losses on others. Reliably selecting context granularity from signals available during training remains an open challenge. Through controlled comparisons of context representations, our study establishes context granularity as a design axis in OPSD. The guiding principle is to evaluate what the teacher sees by what the student learns.

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A. Training and Evaluation Details

A.1. Training Configuration

We specify the exact loss used by our trainer implementation. At each valid completion position, teacher and student logits are divided by the same scoring temperature γ and normalized over the full vocabulary:

$$\bar{q}_{\ell,t} = \text{softmax}(z_{\ell,t}^q/\gamma), \quad \bar{p}_{\theta,t} = \text{softmax}(z_{\theta,t}^p/\gamma). \quad (7)$$

With the configured generalized-divergence endpoint $\beta = 0$ [2], the unreduced vocabulary contribution is

$$u_{\ell,t,v} = \bar{q}_{\ell,t}(v) [\log \bar{q}_{\ell,t}(v) - \log \bar{p}_{\theta,t}(v)]. \quad (8)$$

Before reduction, the implementation applies an upper clamp to each element of the resulting batch–position–vocabulary tensor:

$$\mathcal{L}_{\ell}(\theta) = \frac{1}{T} \sum_{t=1}^T \sum_{v \in \mathcal{V}} \min(u_{\ell,t,v}, \tau). \quad (9)$$

Without the clamp this vocabulary sum would be the teacher-to-student forward KL $D_{\text{KL}}(\bar{q}_{\ell,t} \parallel \bar{p}_{\theta,t})$; with it, the objective is a clipped teacher–student distribution objective rather than an exact KL divergence. Because the clamp precedes the vocabulary sum, it is a vocabulary-component clamp, not a scalar per-token KL clip. The inner sum is the per-position discrepancy $d(q_{\ell,t}, p_{\theta,t})$ of Equation 4, and prompt and padding positions are masked. The clamp thresholds follow the original OPSD configuration for each student scale (Table 4) and are identical across context conditions; τ is distinct from the gradient-norm cap listed in the same table.

Table 4: Training hyperparameters. Scale-dependent values are shown explicitly; all remaining entries are shared.

Hyperparameter	1.7B	4B	8B
Training examples	29,434	29,434	29,434
GPU count	8	8	8
Per-device batch	4	4	2
Gradient accumulation	2	1	2
Effective batch	64	32	32
Optimizer steps	200	200	200
Checkpoint interval	25	25	25
Learning rate	5×10^{-6}	5×10^{-6}	5×10^{-6}
Gradient-norm cap	0.1	0.1	0.1
LoRA rank / scaling	64 / 128	64 / 128	64 / 128
Max. training rollout length	1,024	1,024	1,024
Rollout / scoring temperature	1.1 / 1.1	1.1 / 1.1	1.1 / 1.1
Scoring distribution	Full vocabulary	Full vocabulary	Full vocabulary
Vocabulary-component clip τ	0.05	0.05	0.06

All scales use rollout sampling with temperature 1.1, top- p 0.95, and top- k 20. These truncation parameters affect generation only; the loss in Equation 9 uses the full vocabulary. The training rollout uses Qwen3’s non-thinking chat template, while teacher scoring uses its thinking template; evaluation enables thinking mode. The teacher is the fixed base model with LoRA disabled. All runs use a single node with 8 NVIDIA H200 GPUs.

A.2. Evaluation Protocol

Each AIME24, AIME25, and HMMT25 problem is sampled 12 times at temperature 1.0 with a maximum generation length of 38,912 tokens; Avg@ N denotes accuracy averaged over N sampled completions. The main in-domain table reports the best score observed up to step 200 separately for each (model, condition, benchmark) triple, so an average across three benchmarks can combine scores from different checkpoints. Each unadapted Base is evaluated once under identical decoding and requires no checkpoint selection. The peak protocol takes a maximum over eight correlated evaluations per condition and is therefore upwardly biased relative to any single-checkpoint score; Appendix C.3 quantifies the gap between the best score and the score

at step 200. Differences quoted in the text are computed from unrounded values and may deviate by up to 0.01 points from differences recomputed from rounded table entries.

The transfer suite spans different forms of distribution shift:

- **MT-AIME2024-7Lang** [40]: the same 30 AIME problems translated into Chinese, French, Russian, German, Arabic, Japanese, and Korean. We report the macro average of language-specific Avg@12 scores, weighting languages equally.
- **AutoLogi-EN** [41]: 1,575 open-ended arrangement problems scored by the dataset’s executable verifiers, one generation per problem.
- **GPQA-Diamond** [42]: 198 graduate-level science questions, ten samples per question (Avg@10).
- **ZebraLogic-grid** [43]: 1,000 logic-grid puzzles across 25 grid sizes. The primary metric is exact puzzle accuracy; cell accuracy and difficulty buckets are secondary diagnostics.

C-Eval [44] serves as a broad Chinese knowledge audit under a bounded generation budget; because its result is sensitive to whether a model finishes reasoning within the output budget, it is reported separately from the four reasoning-transfer benchmarks (Table 11).

B. Context Construction and Audits

B.1. Context Generation and a Worked Example

To make the granularity axis concrete, we show all five teacher contexts constructed from the same competition mathematics problem. The student sees only the **Problem** field; during scoring, the teacher receives the complete package $c_\ell(x)$, whose privileged content is $h_\ell(x)$. The boxes below show the hint content; the corresponding bridges are reproduced in Appendix B.2. The example is chosen from the training pool used across all conditions. The stored fields are reproduced faithfully: L2–L5 are verbatim, while L1 preserves every reasoning step with Markdown and minor grammar normalized for typesetting. Its step (1) is labeled “powers of the same base” although the rewriting actually unifies the exponent at 1003; we keep the reference wording as stored, since the conditions are compared on the packages the teacher receives.

Problem x

Let $p = 2^{3009}$, $q = 3^{2006}$, and $r = 5^{1003}$. Which of the following statements is true? (A) $p < q < r$ (B) $p < r < q$ (C) $q < p < r$ (D) $r < p < q$ (E) $q < r < p$

L1: complete reference solution (h_1)

Given the values: $p = 2^{3009}$, $q = 3^{2006}$, $r = 5^{1003}$. (1) Express p and q in terms of powers of the same base: $p = 2^{3009} = 2^{3 \times 1003} = (2^3)^{1003} = 8^{1003}$ and $q = 3^{2006} = 3^{2 \times 1003} = (3^2)^{1003} = 9^{1003}$. Note: $r = 5^{1003}$ is already expressed as a power. (2) Compare the magnitudes: to compare p , q , and r , we now look at the factors 8, 9, and 5, which are 8^{1003} , 9^{1003} , and 5^{1003} . (3) Analyze the base values: base 8 is $8 = 2^3$; base 9 is $9 = 3^2$; base 5 is 5. (4) Since the common exponent in all terms is 1003, we compare the bases: clearly, $5 < 8 < 9$. (5) Therefore, since all exponents are equal, the order of the original terms follows the order of their bases: $5^{1003} < 8^{1003} < 9^{1003}$, so $r < p < q$. Thus, the true statement is $r < p < q$, and the answer is **D**.

L2: named strategy (h_2)

Apply the power-of-a-power rule to rewrite each term with a common exponent equal to the greatest common divisor of the original exponents, enabling direct comparison of the resulting bases.

L3: method-independent framing (h_3)

Reframe the terms into a common representational form so that comparison depends on a single varying attribute.

L4: problem category (h_4)

Elementary algebra: ordering of exponential terms with different bases and exponents.

L5: answer only (h_5)

D

The intended sequence imposes progressively more restrictive roles: relative to L1, L2 omits the executed derivation and answer while retaining a concrete method; L3 omits method identity and retains only a framing shift; L4 retains only the category. Because L2–L4 are separately generated natural-language packages, these relations are operational boundaries, not literal set-theoretic containment. L5 is not on this axis: it retains only the answer.

The staged prompts that enforce these boundaries follow. L2–L4 are generated in three staged calls per problem, $L4 \rightarrow L3 \rightarrow L2$, so that each later stage sees the already generated more abstract levels and can respect the adjacent boundary. Each call receives the problem and the reference solution (truncated to 4,000 characters and marked as “for your understanding only”).

The primary Qwen3.5-397B-A17B compiler runs at temperature 0.3 with a 512-token output cap. The Qwen3-8B replication uses the same prompt texts and staging but enables Qwen3 thinking mode at temperature 0.6, top- p 0.95, and top- k 20; only final response content is retained, with an 8,192-token ceiling to leave room for internal reasoning. Both variants retry up to three times on transport errors or empty output, but never on length. Because model and decoding mode change together, the replication tests a deployable compiler package rather than isolating parameter count.

The prompts impose no hard length constraint; they give a soft “typical range” and explicitly permit deviation, so length varies with the abstraction level and Table 6 reports the realized distributions. Each prompt also contains positive and negative worked examples and mandatory self-check tests that the model must apply before answering. We summarize the binding constraints below; the complete prompt texts, including worked examples, are in the code repository.

L4 (problem category; generated first)

Required pattern: "<Domain>: <noun phrase describing the object / concept type>."; the noun phrase describes WHAT is asked, not how to solve it.
 Must not contain: any solution-action verb (solve, compute, prove, derive, convert, use, apply, calculate, count, find, determine, ...); any method, technique, or theorem name; any thinking pattern or cognitive move; any concrete numerical value or the final answer; any prepositional clause that smuggles in a method ("via X", "using X", "through X", "by X", "based on X", ...).
 Self-checks: verb test; "via X"-removal test (delete the clause; if the remaining sentence no longer classifies the problem, the classification was by method and must be rewritten); method-name test.
 Typical length 10–30 tokens; brevity is a feature of L4, not a flaw.

L3 (method-independent framing; sees L4)

Must be a single cognitive framing shift (e.g., shift to the complement, unify into a common representation, seek an invariant), not an operation sequence.
 Must not contain: any named method, theorem, or algorithm; two or more concrete action verbs (a verb-count test flags an operation sequence in disguise); the words "then", "next", "after", "subsequently"; any concrete numerical value or the final answer.
 Transferability self-check: the same sentence must apply unchanged to a different problem in the same L4 category that requires a different concrete method.
 Typical length 20–60 tokens.

L2 (named strategy; sees L4 and L3)

Must name at least one specific method, theorem, or tool that a student could look up, and may add one short conceptual clause explaining why it applies.
 Must not contain: procedural language ("first X, then Y", numbered steps, "write down", "carry over"); any concrete numerical value from the problem; any symbolic substitution ("let $x = \dots$ "); any intermediate expression or the final answer. "Then" is permitted only to connect two named methods, never two procedural steps.
 Self-checks: sequence-word ban; named-method requirement.
 Typical length 30-90 tokens.

The matched semantic audits of Appendix B.3 evaluate both compilers' realized hints against these definitions on the same 1,200 problems.

B.2. Teacher Interfaces

Every teacher prompt in the primary runs instantiates one fixed template. With {problem} and {hint} denoting the training fields, the teacher-side user message is

```
Problem: {problem}
Here is a {label} for this problem:
{begin delimiter}
{hint}
{end delimiter}
{transition}
```

The student-side user message is identical for all conditions and never contains the hint:

```
Problem: {problem}
Please reason step by step, and put your final answer within \boxed{ }.
```

Table 5 lists the level-specific label and delimiters, and the five transition texts follow. The shared-bridge control replaces all three elements with the single template of Appendix E, and the routed mixtures reuse the level-specific elements below row by row.

Table 5: Level-specific intro label and hint delimiters. The intro line is always "Here is a {label} for this problem:".

Level	Label	Delimiter pair
L1	reference solution	=== Reference Solution Begin/End ===
L2	strategy hint	=== Strategy Hint Begin/End ===
L3	thinking direction	=== Thinking Direction Begin/End ===
L4	problem category	=== Problem Category Begin/End ===
L5	known final answer	=== Final Answer Begin/End ===

All five transitions end with the same closing sentence, quoted once here and elided as [CLOSE] below:

Shared closing, all levels

Think step by step, explore different approaches, and don't be afraid to backtrack or reconsider if something doesn't work out:

The L1-L4 openings are parallel and differ only in the level-appropriate verb chain (*understand-arrive*, *understand-expand*, *grasp-discover*, *recall-select*); L5 instead directs the teacher to construct a derivation toward a given destination.

L1 transition

After reading the reference solution above, make sure you truly understand the reasoning behind each step -- do not copy or paraphrase it. Now, using your own words and independent reasoning, arrive at the correct final answer to the problem above. [CLOSE]

L2 transition

After reading the strategy hint above, make sure you truly understand why this strategy fits the problem -- do not copy or paraphrase it. Now, using your own words and independent reasoning, expand this strategy into a full derivation and arrive at the final answer to the problem above. [CLOSE]

L3 transition

After reading the thinking direction above, make sure you truly grasp how this direction applies to the problem -- do not copy or paraphrase it. Now, using your own words and independent reasoning, discover a concrete method along this direction and arrive at the final answer to the problem above. [CLOSE]

L4 transition

After reading the problem category above, make sure you truly recall the standard techniques used for this type of problem -- do not copy or paraphrase it. Now, using your own words and independent reasoning, select a suitable technique and derive the final answer to the problem above. [CLOSE]

L5 transition

The known final answer above gives only the destination, not the derivation. Do not merely repeat or cite it. Now independently construct a complete, valid reasoning path from the problem to that answer, justifying every necessary step. [CLOSE]

B.3. Context Audits

Table 6 reports exact counts under the Qwen3 tokenizer, which is shared across scales. We report both stored-hint length and the privileged increment, defined as the teacher prompt minus its matched no-hint prompt, because the fixed bridge is a substantial fraction of short contexts. All levels contain 29,434 aligned, nonempty rows with no error sentinels; L5 includes seven proof-target overrides. These reductions concern the teacher-scoring interface once a solution exists, not solution acquisition or end-to-end wall-clock cost.

Table 6: Exact Qwen3-token lengths over all 29,434 training examples. P10 and P90 apply to raw hint content; the last two columns are means. Privileged increment includes the hint and its level-specific bridge but excludes the problem and shared chat-template tokens.

Context	Hint mean	Median	P10	P90	Teacher prompt	Privileged increment
L1: solution	657.3	630	322	1,022	835.0	734.4
L2: strategy	38.9	38	31	48	225.5	124.9
L3: framing	25.2	25	20	32	212.7	112.2
L4: category	17.7	18	14	22	203.3	102.7
L5: answer/target	6.7	4	1	15	183.9	83.4

The deterministic scan covers all 29,434 rows from each compiler. For semantic properties, we draw 1,200 unique problem rows with seed 20260719, proportionally stratified over four answer types (numeric, expression or text, multiple choice, and proof target) and four reference-solution length quartiles. The same rows are used for both compilers, each contributing 3,600 audited hints (1,200 problems $\times$ three levels). The audit distinguishes *direct leakage*, which explicitly states the final value, expression, option, or answer, from *answer-equivalent leakage*, which states a conclusion that determines the final answer without carrying out the

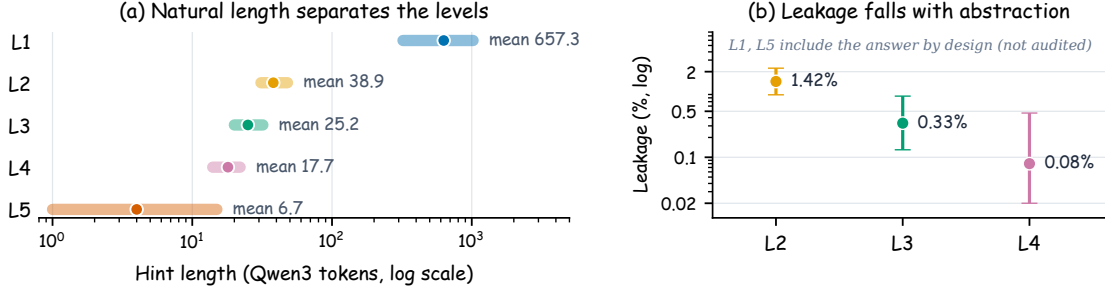


Figure 4: Primary-compiler context length and answer-leakage audit. **(a)** Hint length across all 29,434 problems, measured in Qwen3 tokens and shown on a log scale; Table 6 reports summary statistics. **(b)** Adjudicated answer-equivalent leakage on a stratified sample of 1,200 problems. L1 and L5 contain the answer by design; Table 7 reports matched audits of both compilers.

intended solution. A useful method alone is not counted as leakage. Boundary compliance additionally requires L2 to name a concrete method without executing a derivation, L3 to provide one method-independent framing without an operation sequence, and L4 to identify the correct domain and objects without recommending an action.

Two zero-temperature prompt variants independently annotate every compiler–problem triplet. Of the triplets with at least one disagreement, we send the 31 from the primary compiler and the 67 from the Qwen3-8B compiler to a third adjudication pass using the same inputs without access to the earlier judgments. This resolves all 2,400 compiler–problem records, with no unresolved cases. All three passes use Qwen3.5-397B-A17B, so the resulting agreement is prompt-replication agreement from one model rather than human inter-rater agreement.

Table 7: Matched semantic audits of both compilers on the same 1,200 problem rows after adjudication. Rates are computed over the 1,200 hints at each compiler–level pair. “Core role valid” checks concrete-method presence for L2, single-framing presence for L3, and both domain and object correctness for L4.

Compiler	Level	Direct	Answer-equivalent	Boundary pass	Core role valid
397B primary	L2: strategy	0/1200	17/1200 (1.42%)	98.58%	100.00%
	L3: framing	0/1200	4/1200 (0.33%)	99.67%	100.00%
	L4: category	0/1200	1/1200 (0.08%)	99.92%	100.00%
8B replication	L2: strategy	0/1200	50/1200 (4.17%)	95.58%	99.92%
	L3: framing	0/1200	31/1200 (2.58%)	97.08%	99.83%
	L4: category	0/1200	2/1200 (0.17%)	98.83%	99.17%

No hint directly copies an answer. The 8B compiler has higher L2/L3 leakage and boundary-failure rates, whereas its L4 context remains low-leakage and follows the scale-dependent direction in Table 12. The L4 result shows that the finding is not confined to the leakier L2/L3 contexts; leakage may still affect individual L2/L3 cells.

Table 8 reports agreement for the two fields with positive adjudicated cases.

The audit release accompanying the code will include sample IDs, prompts, raw outputs, adjudications, and aggregation code.

The two replicated audit passes and the adjudication pass share one block of operational definitions and differ only in the role preamble:

(A) You are a conservative data-quality auditor for mathematical reasoning supervision. Apply the operational definitions literally and do not infer unstated content.

(B) You are an adversarial compliance reviewer. Look independently for subtle answer-equivalent leakage and boundary drift, but do not penalize a hint merely because it is useful. Evaluate

Table 8: Prompt-A/B agreement on the two audit fields that contain positive adjudicated cases. For primary-compiler L4, $\kappa = 0$ is a prevalence artifact: one prompt assigns no positives and the other assigns only two, despite 99.83% raw agreement.

Level	Field	397B primary		8B replication	
		Agreement	κ	Agreement	κ
L2	Answer leakage	97.83%	0.527	96.75%	0.695
L2	Boundary pass	97.67%	0.507	96.33%	0.680
L3	Answer leakage	99.58%	0.614	98.17%	0.694
L3	Boundary pass	99.58%	0.614	97.92%	0.673
L4	Answer leakage	99.83%	0.000	100.00%	1.000
L4	Boundary pass	99.83%	0.000	99.33%	0.711

each level against its absolute definition, not relative to the other hints.

(C) You are the final independent adjudicator for a mathematical hint audit. Resolve the case from the source text alone; you are not shown earlier judgments. Be strict about evidence and use ‘uncertain’ only when the source wording truly admits two readings.

The shared operational definitions are:

Audit the three hints against their declared definitions. Judge only what is actually written in each hint, using the problem, reference solution, and ground-truth answer only as verification context.

ANSWER LEAKAGE:

- "direct": explicitly states the final value, expression, option, or answer.
- "equivalent": states a conclusion algebraically/logically equivalent to the final answer, so the answer is already determined without carrying out the intended solution.
- "none": contains only guidance or classification. A useful strategy that makes the problem easier is NOT leakage.
- "uncertain": evidence is genuinely ambiguous.

L2 (named strategy): Must identify a concrete method, theorem, rule, representation, or tool. May briefly say why it fits. Must not carry out problem-specific calculations or a derivation. Must not give a multi-step operation sequence or the answer.

L3 (method-independent framing): Must express one transferable cognitive framing shift. Must not name a specific theorem, algorithm, or concrete method. Must not prescribe a sequence of operations or give the answer.

L4 (problem category): Must correctly identify the mathematical domain and relevant object/concept. Must not recommend a method, cognitive move, action sequence, or answer.

For every false/violation judgment, quote at most two short exact spans from the corresponding hint in "evidence". Do not quote the reference solution. Set boundary_pass from the complete definition, not merely from answer leakage. Return JSON only, with exactly the requested schema.

The user message supplies the sample ID, problem, ground-truth answer, the reference solution (as verification context, clipped to 24,000 characters), the paired L2/L3/L4 hints, and the required JSON schema. Each level is annotated with `answer_leakage` $\in$ {none, direct, equivalent, uncertain}, level-specific boolean boundary fields (L2: `concrete_method_present`, `problem_specific_execution`, `multi_step_sequence`; L3: `single_framing_present`, `named_method_present`, `operation_sequence`; L4: `domain_correct`, `objects_correct`, `method_or_action_present`), an overall `boundary_pass`, a confidence grade, and at most two verbatim evidence spans of at most 240 characters. Responses that violate this schema are rejected and retried, so every retained annotation is structurally valid.

C. Full Results and Robustness

C.1. Full Evaluation Results

Table 9 gives the complete transfer results at step 200 summarized in Section 4.1, and Table 10 expands ZebraLogic into cell accuracy and the Hard and XL subsets. Both are single-run comparisons at a common endpoint; the multi-seed analysis covers the L1, L3, and L4 in-domain comparison (Appendix C.3). Table 11 reports the separately interpreted C-Eval audit, discussed at the end of this subsection.

Table 9: Transfer using the fixed checkpoint at step 200, with the primary compiler for L2–L4. Scores are percentages: MT-AIME reports macro Avg@12 over seven languages, GPQA Avg@10, AutoLogi verifier accuracy, and ZebraLogic exact puzzle accuracy. L5 is the answer-only control. Shading marks the highest and second-highest point estimates within each scale and benchmark, including ties.

Model	Condition	MT-AIME-7Lang	AutoLogi	GPQA-Diamond	ZebraLogic
1.7B	Base	42.62	86.73	36.72	61.20
	L1	46.94	87.43	38.54	65.10
	L2	46.79	86.22	37.98	62.60
	L3	48.65	86.54	39.34	64.40
	L4	47.86	87.62	37.73	62.70
	L5	44.84	86.54	38.18	61.30
4B	Base	65.87	91.75	53.69	82.20
	L1	67.46	92.38	54.75	82.50
	L2	68.85	92.13	55.96	82.10
	L3	67.62	92.63	55.25	82.60
	L4	69.56	92.25	55.51	82.90
	L5	68.21	92.76	53.28	82.40
8B	Base	70.48	92.13	60.05	85.60
	L1	71.90	92.63	61.26	86.40
	L2	73.61	92.13	60.45	87.30
	L3	72.70	91.87	62.68	88.10
	L4	72.74	92.13	60.51	87.60
	L5	71.43	91.94	61.67	85.90

C-Eval is a bounded-generation audit rather than a reasoning-transfer benchmark, so it is read on its own. Under a 32,768-token output budget, no training condition improves on the unadapted Base consistently across scales on C-Eval Macro (Table 11). Base is the best condition at 1.7B (69.82 against 69.23 for L1) and at 8B (84.60 against 84.57 for L5); the one condition that exceeds Base anywhere is the answer-only control L5 at 4B, by 0.07 points (80.80 against 80.73), while none of L1–L4 exceeds Base at any scale. Extraction failures rise under every training condition, most sharply at 1.7B, from 0.68% for Base to 3.70% for L2, and the metric charges each failure as an error. Under this protocol, therefore, OPSD training on competition mathematics yields no consistent gain on broad Chinese knowledge questions, and it leaves more responses without a parseable answer. This is why C-Eval is kept apart from the four reasoning-transfer benchmarks in the main text.

C.2. Context Controls

Two controls vary how the privileged context is produced while holding the training pipeline fixed. Table 12 regenerates L2–L4 with Qwen3-8B under the same semantic contracts, so the ordering between full and intermediate contexts can be checked against a second compiler. Table 13 replaces semantic rewriting with a fixed 10% reference prefix plus the answer at 4B, which shortens the context without abstracting it; that comparison matches L2–L4 in neither length nor answer access. Section 4.2 discusses both results.

Table 10: Complete ZebraLogic-grid results at step 200. Puzzle is exact-grid accuracy; Cell measures individual grid entries. Hard and XL are exact-puzzle accuracy on the corresponding subsets. Δ from Base is the absolute percentage-point change from the same-scale unadapted Base checkpoint. Light-blue/bold shading marks the best and light-gray/underline shading the second-best condition within each model scale.

Model	Condition	Puzzle	Δ from Base	Cell	Hard	XL
1.7B	Base	61.20	–	68.88	48.06	7.00
	L1	65.10	+3.90	<u>72.74</u>	53.47	10.50
	L2	62.60	+1.40	70.41	49.72	8.00
	L3	<u>64.40</u>	<u>+3.20</u>	73.09	<u>53.06</u>	7.50
	L4	62.70	+1.50	71.48	50.42	<u>9.00</u>
	L5	61.30	+0.10	69.84	47.92	6.00
4B	Base	82.20	–	84.21	75.42	<u>36.50</u>
	L1	82.50	+0.30	84.83	75.97	36.00
	L2	82.10	–0.10	83.14	75.42	34.00
	L3	<u>82.60</u>	<u>+0.40</u>	<u>85.38</u>	<u>76.25</u>	36.00
	L4	82.90	+0.70	85.91	76.53	36.00
	L5	82.40	+0.20	84.78	75.69	37.50
8B	Base	85.60	–	86.80	80.14	41.00
	L1	86.40	+0.80	86.85	81.53	44.50
	L2	87.30	+1.70	87.45	82.64	46.00
	L3	88.10	+2.50	88.25	83.89	<u>46.50</u>
	L4	<u>87.60</u>	<u>+2.00</u>	<u>87.83</u>	<u>83.06</u>	47.50
	L5	85.90	+0.30	86.13	80.97	43.50

Table 11: C-Eval test results with a 32,768-token output budget. Macro is the mean over 52 subjects; Hard Macro covers the official C-Eval Hard subjects. Extraction failures are counted as incorrect in Macro and Micro. Light-blue/bold shading marks the best and light-gray/underline shading the second-best condition within each model scale.

Model	Condition	Macro	Micro	Hard Macro	Extraction fail.
1.7B	Base	69.82	67.97	73.77	0.68
	L1	<u>69.23</u>	<u>67.49</u>	73.19	<u>1.67</u>
	L2	67.46	65.78	72.31	3.70
	L3	67.86	66.25	73.35	3.04
	L4	67.78	66.11	<u>73.75</u>	3.48
	L5	68.49	66.85	72.68	2.43
4B	Base	<u>80.73</u>	79.40	<u>85.91</u>	0.77
	L1	80.03	78.73	84.96	2.26
	L2	80.11	78.78	84.45	2.31
	L3	80.09	78.76	85.21	2.21
	L4	80.52	79.14	85.18	<u>1.77</u>
	L5	80.80	<u>79.37</u>	86.28	1.96
8B	Base	84.60	83.61	88.45	0.84
	L1	84.43	83.37	<u>88.47</u>	1.27
	L2	84.13	82.98	87.75	1.43
	L3	84.31	83.26	88.09	1.51
	L4	84.53	<u>83.46</u>	87.89	1.26
	L5	<u>84.57</u>	83.41	89.40	<u>1.17</u>

Table 12: Compiler replication using ID peak means. L2–L4 are generated with the primary Qwen3.5-397B-A17B compiler or with Qwen3-8B; L1 requires no compiler and is the shared reference at each scale for both halves, not an additional run. Bold marks the best condition within each compiler block, counting the shared L1 column. The peak ordering between full and intermediate contexts is preserved across compilers, while the preferred intermediate level can change.

Student	L1	397B primary compiler			Qwen3-8B compiler		
		L2	L3	L4	L2	L3	L4
1.7B	43.89	40.83	42.78	43.61	41.02	42.78	42.96
4B	63.43	64.07	64.81	64.63	64.07	63.71	64.35
8B	66.67	68.24	66.85	67.22	67.96	67.04	67.13

Table 13: Comparison of semantic contexts and prefix truncation with Qwen3-4B. ID peak mean averages each benchmark’s best Avg@12 across eight checkpoints, while ID@200 uses step 200. Only the prefix condition is new; the L1–L5 rows repeat the 4B primary runs, whose per-benchmark peaks are in Table 1. The upper block includes the final answer, the lower block prohibits it by design. The comparison matches neither length nor answer access.

Condition	ID peak mean	ID@200
L1: complete solution	63.43	60.93
L5: answer only	63.61	62.50
Prefix-10% + answer	63.89	61.67
L2: strategy	64.07	62.59
L3: framing	64.81	61.76
L4: category	64.63	62.59

C.3. Training Robustness

Three checks probe whether the granularity contrast depends on the training budget, the optimization seed, or the checkpoint at which runs are read.

The primary experiments fix the training rollout budget at 1,024 tokens, following the original OPSD configuration. Because sampled completions frequently reach this cap, we verify that the granularity comparison survives a doubled budget: L1–L4 are retrained at 4B with a 2,048-token rollout budget and otherwise identical configuration, then evaluated under the standard in-domain protocol.

Table 14: Robustness to rollout budget at 4B (Avg@12 means across three benchmarks). Peak takes each benchmark’s best checkpoint; single-ckpt selects the one checkpoint with the best mean across three benchmarks (step in parentheses); @200 is the fixed endpoint. Truncation reports the fraction of training rollouts reaching the token cap early (steps ≤ 50) and late (steps > 150) in training. Bold marks the column maximum, including ties. Intermediate levels match or beat L1 on the peak and single-checkpoint means under both budgets.

Level	1,024-token rollouts				2,048-token rollouts			
	Peak	Single-ckpt	@200	Trunc. e/l	Peak	Single-ckpt	@200	Trunc. e/l
L1	63.43	62.31 (125)	60.93	57%/77%	63.43	63.06 (50)	60.93	21%/58%
L2	64.07	63.43 (125)	62.59	56%/83%	64.54	64.26 (75)	58.61	21%/67%
L3	64.81	63.24 (150)	61.76	57%/81%	64.35	63.70 (175)	61.48	22%/52%
L4	64.63	64.63 (125)	62.59	55%/77%	65.37	64.63 (75)	61.57	22%/54%

Table 14 shows that the doubled budget roughly halves truncation early in training, yet changes each level’s peak mean by at most 0.74 points. Every intermediate context again matches or exceeds L1, while the nominal leader shifts from L3 to L4. The larger budget moves three of four single-checkpoint peaks earlier and does not improve any endpoint at step 200, suggesting a change in effective training horizon while preserving the family-level ordering. The endpoint is more sensitive to the budget: L2’s mean at step 200 falls from 62.59 to

58.61 even though its peak mean is higher under the larger budget (64.07 to 64.54), so preserving the peak does not imply preserving the endpoint. These runs use one seed at 4B and do not evaluate transfer.

To probe optimization variance directly, L1, L3, and L4 are retrained at all three scales with two additional configurations under Table 4. The primary run uses the default seed 42; the additional runs use seeds 43 and 44, with the trainer, data order, and rollout RNGs changed together and disjoint across configurations. Every run is evaluated with the standard in-domain protocol. Table 15 groups the per-benchmark bests, the peak mean across three benchmarks, and the endpoint at step 200 by configuration, so each block compares runs with the same seed configuration, and reports the paired peak difference from that configuration’s own L1 run.

Table 15: Replication across training seeds at all three scales (Avg@12). ID peak takes the best checkpoint per benchmark up to step 200 and averages the three; ID@200 is the selection-free endpoint mean. Rows are grouped by seed configuration, so every block compares intermediate contexts against its own L1 run; seed 42 is the primary run of Table 1, and 43 and 44 are the two additional configurations. Δ is the ID peak difference from the L1 run in the same block. The best scores are not shaded because the configurations do not agree on an intermediate leader.

Student	Seed	Level	AIME24	AIME25	HMMT25	ID peak	Δ vs. L1	ID@200	
1.7B	42	L1	57.22	43.33	31.11	43.89	—	42.31	
		L3	56.94	41.39	30.00	42.78	−1.11	40.00	
		L4	59.72	43.06	28.06	43.61	−0.28	39.26	
	43	L1	55.28	43.89	28.61	42.59	—	38.89	
		L3	57.22	41.67	30.00	42.96	+0.37	40.74	
		L4	58.06	42.22	28.61	42.96	+0.37	39.81	
	44	L1	56.39	41.39	29.72	42.50	—	41.11	
		L3	58.33	41.39	28.89	42.87	+0.37	39.07	
		L4	57.50	41.94	27.78	42.41	−0.09	39.26	
	4B	42	L1	76.94	68.61	44.72	63.43	—	60.93
			L3	77.50	69.72	47.22	64.81	+1.39	61.76
			L4	76.94	69.17	47.78	64.63	+1.20	62.59
43		L1	76.67	69.44	45.28	63.80	—	60.83	
		L3	76.39	68.89	46.11	63.80	0.00	63.06	
		L4	76.94	70.28	46.67	64.63	+0.83	62.78	
44		L1	76.11	68.06	46.11	63.43	—	61.39	
		L3	76.67	70.00	45.83	64.17	+0.74	63.33	
		L4	76.94	69.72	46.11	64.26	+0.83	60.93	
8B		42	L1	78.89	73.06	48.06	66.67	—	65.28
			L3	79.44	71.39	49.72	66.85	+0.19	65.37
			L4	80.56	71.39	49.72	67.22	+0.56	64.63
	43	L1	78.61	73.06	47.78	66.48	—	63.61	
		L3	78.89	73.61	49.72	67.41	+0.93	63.70	
		L4	80.56	73.06	48.33	67.31	+0.83	65.00	
	44	L1	79.17	73.06	48.33	66.85	—	65.28	
		L3	79.17	72.22	48.06	66.48	−0.37	65.93	
		L4	79.72	71.94	49.17	66.94	+0.09	65.19	

At 4B and 8B, L4 exceeds its matched L1 peak in all three configurations and L3 does so in four of the six scale–configuration cells, falling level at 4B seed 43 and below at 8B seed 44 (Table 15). Aggregating these runs under the four checkpoint rules of Table 3 preserves the direction at both scales, including when all checkpoints are averaged. At 1.7B the peak differences range from −1.11 to +0.37 points and the selection-free rules favor L1 on average, by 0.14 to 1.33 points.

The endpoint at step 200 is the most sensitive to seed among the four rules, carrying the largest paired standard deviation in five of the six contrasts; the exception is 8B L3, whose peak spread of 0.65 exceeds its endpoint spread of 0.32. Among L3 and L4, the leader is not stable within any scale: at 4B, L3 leads the primary run and L4 leads seeds 43 and 44; at 8B, L4 leads the primary run and seed 44 while L3 leads seed 43. These

replications support the contrast between L1 and intermediate contexts at the larger scales, while the ordering among intermediate levels remains unstable. They cover L1, L3, and L4; L2, L5, the transfer evaluations, and the remaining controls are single runs.

Figure 6 shows in-domain Avg@12 at every checkpoint (steps 25, 50, ..., 200) on AIME24, AIME25, and HMMT25, with open circles at each curve’s best checkpoint. The benchmark curves are visibly nonmonotonic, which is what motivates reporting peaks and selection-free endpoints side by side. Table 16 lists the step at which each peak in Table 1 occurs. The selected steps are spread across the schedule rather than concentrated at its end, which is the same observation from the tabular side.

Table 16: Step at which each peak Avg@12 in Table 1 is attained, primary run (seed 42). Checkpoints are taken every 25 steps up to 200.

Model	Benchmark	L1	L2	L3	L4	L5
1.7B	AIME24	125	125	75	75	50
	AIME25	200	75	25	125	100
	HMMT25	100	150	150	75	200
4B	AIME24	25	50	100	125	100
	AIME25	100	150	50	125	75
	HMMT25	175	100	125	125	75
8B	AIME24	175	175	150	25	50
	AIME25	125	100	125	25	150
	HMMT25	75	150	150	25	200

Figure 5 shows the logged clipped teacher–student objective as an auxiliary trace. Each condition optimizes against a different teacher target on different on-policy rollouts, so its values are not on a common scale across conditions and cannot be read as evidence that the conditions trained equally well; the curves only indicate that no run diverged.

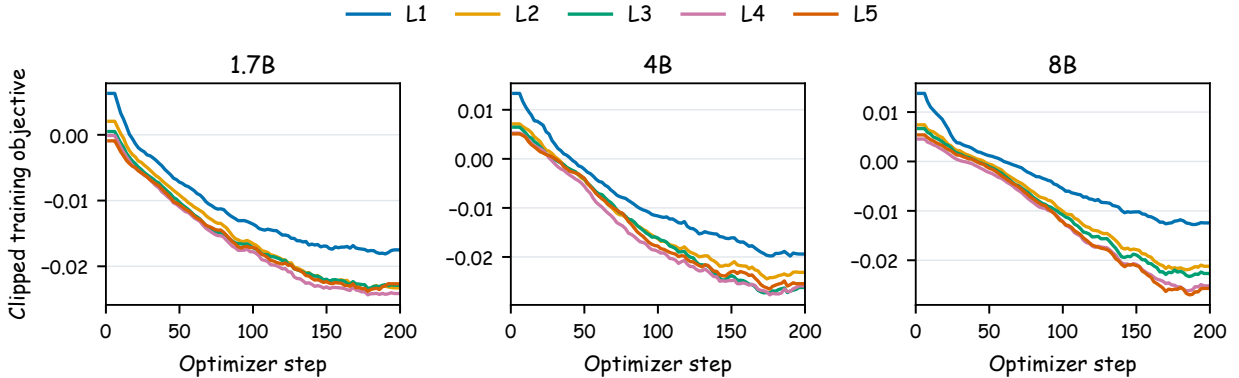


Figure 5: Logged clipped training objective for each teacher context condition and model scale, under a moving average over five records. Each condition has a different teacher target, so the vertical ordering describes the respective objectives only and is comparable neither across conditions nor to the downstream benchmark ranking.

For each (model, condition, benchmark) triple, define $\Delta_{200} = \text{Avg@12}_{\text{best} \leq 200} - \text{Avg@12}_{200}$ in percentage points. Table 17 averages this quantity over the three in-domain benchmarks. Across all 45 triples, the median and mean gaps are 1.94 and 2.31 points, respectively. The largest is 5.83 points for 1.7B-L4 on AIME25 (43.06 at step 125 versus 37.22 at step 200). Thus, step 200 is a common, selection-free endpoint and need not coincide with convergence. The OOD comparison remains internally matched because every condition uses the same stopping rule; sensitivity to alternative fixed endpoints remains open.

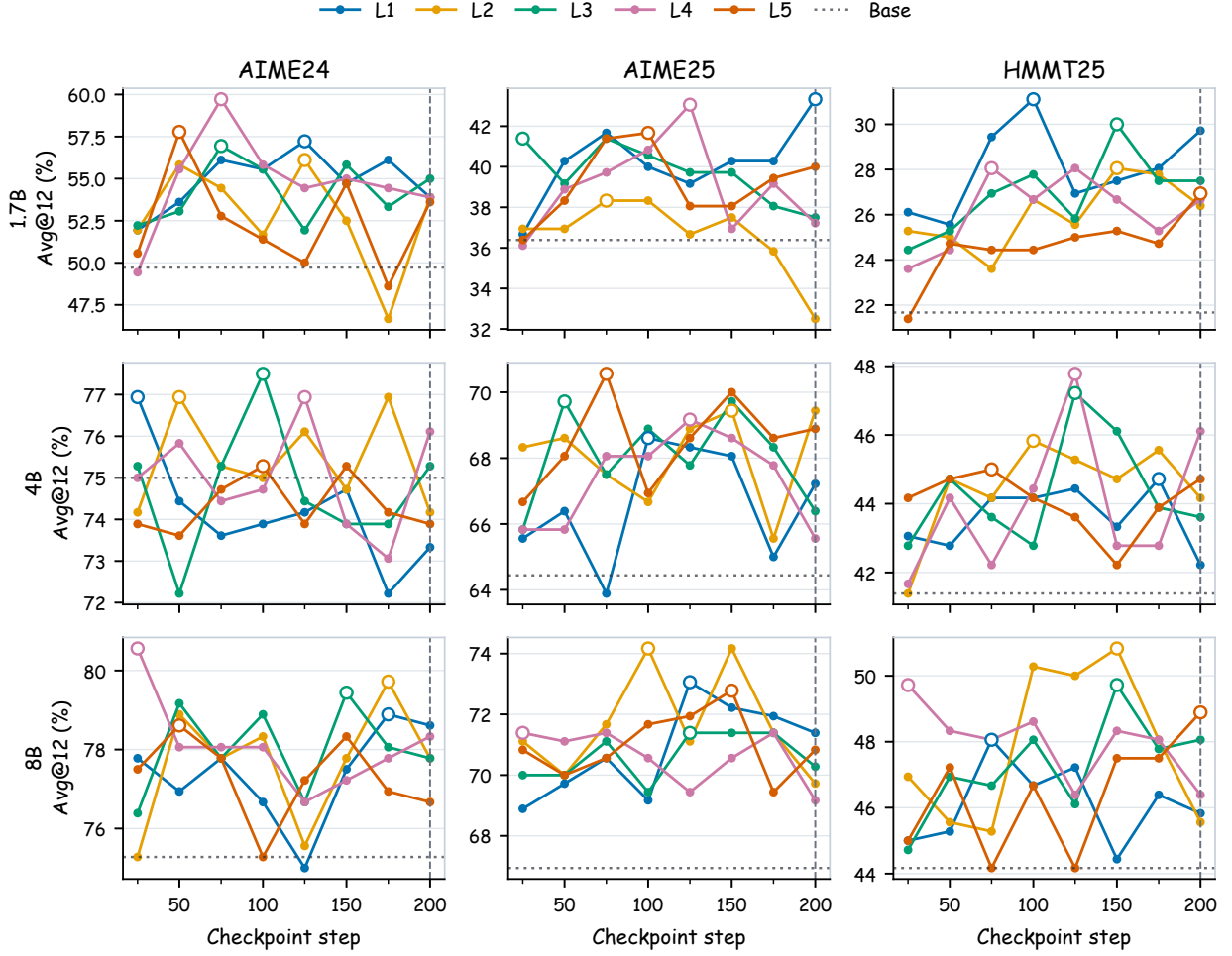


Figure 6: In-domain Avg@12 vs. checkpoint step for each condition and benchmark, primary run (seed 42), with the unadapted Base score shown for reference. Open circles mark each curve’s maximum through step 200; the dashed vertical line marks the common endpoint used for transfer evaluation. The substantial variation across checkpoints motivates separating in-domain peak reporting from transfer at a fixed checkpoint.

Table 17: Mean Avg@12 gap between the best checkpoint through step 200 and the endpoint at step 200 (Δ_{200} , percentage points), averaged over AIME24, AIME25, and HMMT25. Smaller is closer to the condition’s best checkpoint.

Model	L1	L2	L3	L4	L5
1.7B	1.57	3.24	2.78	4.35	1.95
4B	2.50	1.48	3.06	2.04	1.11
8B	1.39	3.89	1.48	2.59	1.30

C.4. Scope of the Controls

This subsection records which controls bound the scope conditions summarized in Section 4.5. Against single-run evidence: replication across three seeds covers L1, L3, and L4 at all three scales (Appendix C.3), while the prefix, bridge, and rollout budget controls center on 4B and the mixture and routing studies are single runs. Against selection by score: the in-domain protocol picks the best of eight checkpoints separately per benchmark, so we report selection-free endpoints and full trajectories in Appendix C.3.

Against bridge wording, which varies with level in the primary runs: the shared-bridge control holds it fixed (Appendix E). Against the confound between semantics and length, which the levels vary together: the prefix control fixes truncation with answer access, though it matches L2–L4 in neither length nor answer access (Appendix C.2). Against compiler identity: the Qwen3-8B replication stays inside the Qwen family and shows higher L2/L3 leakage, and both compilers receive the reference solution, so it broadens the compiler evidence without isolating parameter count.

Against difficulty-bucket artifacts: grouping by Base rather than L1 success rates tests the stratification reference (Section 4.4), but selecting the intermediate condition on the same evaluation used for the difficulty analysis remains a source of selection bias. One model judge supplies both the semantic audits and the trajectory annotations, so replicated prompts bound prompt sensitivity but not judge independence.

Broader validation should extend the comparison to other model families and domains, disentangle semantics from length, and study compilers that construct useful contexts without reference solutions. The ordering across scales motivates studying whether useful context changes as the same student improves. In-domain scores do not improve monotonically across checkpoints, so multi-round OPSD with online context selection and stability-aware optimization is a natural next setting. The hierarchy introduced here provides a controlled interface for such work.

D. Diagnostic Analyses

D.1. Distributional Discrepancy

We deterministically sample 600 aligned training problems with seed 20260804. For each scale, the unadapted model produces two continuations per problem from the exact OPSD student prompt with thinking disabled, using temperature 1.1, top- p 0.95, top- k 20, and at most 1,024 tokens. Each continuation is held fixed while the same base checkpoint scores it under the student prompt, a teacher prompt with thinking enabled but no hint, and the exact L1–L5 teacher packages. Logits are normalized over the full vocabulary at temperature 1.1; prompts are not truncated. We average token-level measurements within each rollout and divide each continuation into four equal bins by relative position. The analyses of whole trajectories and quartiles use all 3,600 scored rollouts, with 1,200 per scale. Bootstrap intervals in the diagnostic figures resample problems together with their associated rollouts, and therefore quantify evaluation uncertainty conditional on the evaluated runs rather than optimization variability.

For the follow-up temporal diagnostic, we locate the last `\boxed` marker by exact decode–re-encode token offsets. We define a pre-answer window from up to 32 tokens immediately before the marker and an answer-onward window from the marker through at most 32 tokens, requiring at least eight tokens in each window. This yields 610, 636, and 630 valid rollouts at 1.7B, 4B, and 8B, spanning 358, 371, and 364 problems. The 1,024-token generation limit truncates 505, 488, and 524 of the 1,200 rollouts at the respective scales (40.7–43.7%); most truncated trajectories never reach a boxed answer and are therefore absent from the aligned subset. The subset consequently favors completed trajectories containing a boxed answer.

As an outcome-conditioned check, we repeat the paired answer-onward contrast on the 421, 510, and 499 valid rollouts whose boxed answer matches the reference under a normalized answer-string heuristic. The resulting L5 – L1 contrasts are -0.193 , $+0.030$, and $+0.188$, preserving the scale-dependent sign change. This normalized answer-string check confirms that the sign change persists within apparently correct rollouts; it remains an outcome-composition diagnostic. Figure 7 reports the corresponding quartile-resolved displacement curves.

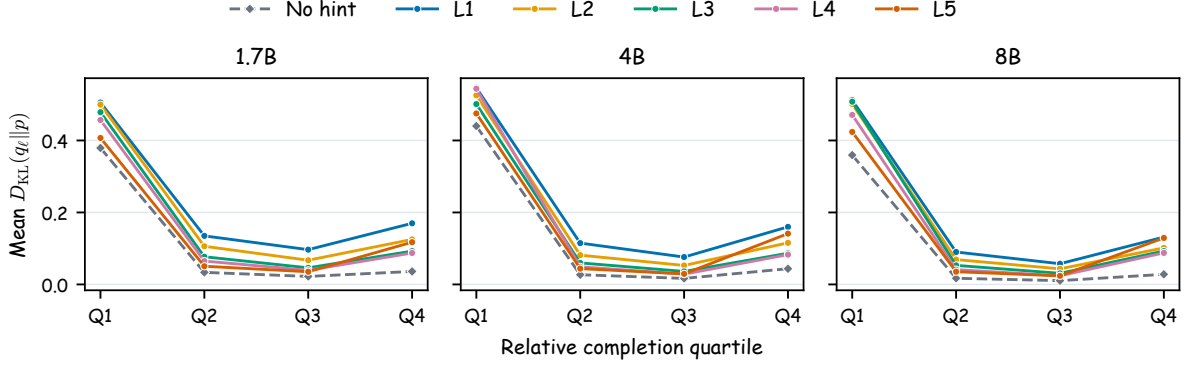


Figure 7: Temporal geometry at each scale, using rollouts from the exact OPSD student prompt. Curves report mean unclipped full-vocabulary teacher-to-student KL within quartiles of relative completion position, using all 1,200 rollouts at each scale; Q4 is the final quarter of each rollout, not the answer region, and this analysis therefore avoids the boxed-answer filtering above without eliminating truncation bias. The no-hint reference uses the thinking-enabled teacher interface without privileged content. L5 has the smallest mean displacement among L1–L5 through Q2, remains among the smallest in Q3, and has the largest rebound from Q3 to Q4 at every scale, both features that means over whole trajectories hide.

D.2. Item-Level and Behavioral Analysis

The frozen manifest for behavioral analysis contains 120 runs evaluated at step 200 and 55,134 problem-condition records, with eight shards per run and no failures in the deterministic artifact analysis. The manifest predates the later addition of the 4B/8B in-domain Base evaluations, which are used in the main score tables but are not required by the behavioral contrasts between L1 and intermediate contexts. For sampled-answer tasks, per-generation accuracy (*reliability*) is complemented by coverage (at least one correct generation) and majority-vote consensus; MT-AIME is grouped by source problem for cross-language analysis, and for ZebraLogic the gap between cell and exact-puzzle accuracy measures whether locally correct constraints close into a complete grid. Response length uses a tokenizer-free word/punctuation proxy and is compared only within a benchmark.

For the model-judged behaviors, the frozen manifest requests four generations for each of 30 matched AIME24 problems and 60 matched GPQA problems per scale and available condition. After structural validation, behavior rates use up to four valid generations per problem (5,997 of 6,000 requested annotations overall). Trajectories longer than 24,000 characters retain their beginning and end. The manifest stratifies problems by the reference present when it was frozen: four of the six scale-benchmark groups use Base, while 4B and 8B AIME24 use L1 because their Base artifacts were added later. This stratification only balances problem selection; the reported behavior rates pool across difficulty bins. The judge additionally labels method family, method switching, and answer-before-justification beyond the fields discussed in the main text.

The trajectory judge used in Section 4.4 follows one zero-temperature system prompt:

```
You are a careful annotator of mathematical and logical reasoning trajectories. Judge only
the behavior visible in the trajectory. Do not reward verbosity and do not infer hidden
intent. Return one JSON object with exactly these fields: {primary_method, secondary_method,
method_switch_count, verification_present, backtracking_present, explicit_error_correction,
successful_recovery, answer_before_justification, premature_commitment, reasoning_complete,
brief_evidence}
```

Definitions:

- A method switch requires abandoning or materially replacing an approach, not routine algebra.
- successful_recovery requires a visible mistake or rejected path followed by a valid recovery.
- answer_before_justification means a candidate/final answer is stated before its supporting derivation.
- premature_commitment means the trajectory commits early and fails to adequately test the

```
commitment.
- Normalize primary_method labels, e.g. algebra, geometry, counting, number_theory,
  case_analysis, contradiction, constraint_propagation, elimination, scientific_knowledge,
  estimation, other.
```

The user message contains the problem, the ground-truth answer, the model’s extracted final answer, and the trajectory (truncated as described above). The judge runs at temperature zero with a 500-token response budget, and structurally invalid JSON responses are retried.

These floats support the item-level and behavioral analyses of Section 4.4. Table 18 gives the difficulty-conditioned changes; because its buckets are defined by L1’s own per-item success rate, part of the easy-bucket decline and the unsolved-bucket recovery is expected from regression to the mean, which is why Section 4.4 repeats the grouping with Base success rates. Figure 8 gives the ZebraLogic rescue-versus-regression counts, and Figure 9 separates MT-AIME per-generation reliability from Pass@12 coverage. Table 19 reports the per-scale judged behavior rates and Figure 10 the pooled differences with bootstrap intervals.

Table 18: Difficulty-conditioned redistribution relative to L1 at step 200. Difficulty buckets use L1’s per-item success rate with the thresholds defined in Section 4.4. New solve is the fraction of L1-unsolved evaluation instances for which the comparison condition produces at least one correct sample. Medium and easy report changes in per-generation correctness. MT-AIME difficulty rows treat each translated instance separately.

Model	Benchmark	Contrast	Overall Δ	New solve	Medium Δ	Easy Δ
1.7B	MT-AIME	L3 – L1	+1.71	12.3	+5.65	−4.68
4B	MT-AIME	L4 – L1	+2.10	16.7	+8.87	+0.68
8B	MT-AIME	L2 – L1	+1.71	21.4	+15.79	−0.11
1.7B	GPQA	L3 – L1	+0.81	24.2	+0.67	−3.70
4B	GPQA	L2 – L1	+1.21	25.0	+2.63	−1.12
8B	GPQA	L3 – L1	+1.41	25.6	+5.26	−0.96
1.7B	ZebraLogic	L3 – L1	−0.70	18.1	−	−10.75
4B	ZebraLogic	L4 – L1	+0.40	29.1	−	−5.70
8B	ZebraLogic	L3 – L1	+1.70	35.3	−	−3.59

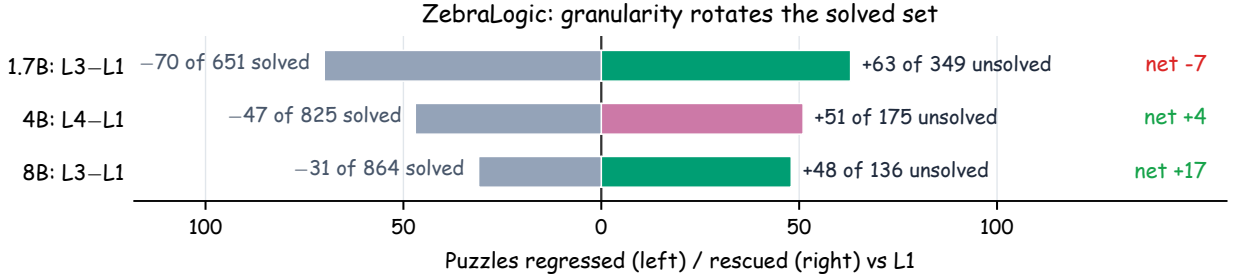


Figure 8: Whether rotation pays depends on capacity. For each scale, the bars count ZebraLogic puzzles the strongest intermediate condition rescues from L1-unsolved (right, level color) and gives back from L1-solved (left, gray) at step 200. The rescue rate rises with scale (18% to 35% of L1-unsolved puzzles) while regressions shrink, turning a net loss at 1.7B into a net gain of +17 puzzles at 8B.

Length does not move in one direction with the gains: selected intermediate responses are usually 4–10% longer than L1 on MT-AIME and GPQA but 3.7% shorter on 8B ZebraLogic. A uniform length effect is therefore ruled out, though length may still contribute on individual tasks.

One contrast case makes the cell-to-puzzle gap concrete. To keep the choice independent of the narrative, we show one case from the predeclared contrast set (winner per-item success ≥ 0.75 , loser ≤ 0.25 , selected before reading any trajectory). The case is ZebraLogic puzzle `lgp-test-4x6-25` at 8B: four houses with six attribute categories. L3 solves the grid in a 33,912-character trajectory that terminates normally; L1 produces a 67,938-character trajectory, also terminating normally, whose final grid is wrong. Both models make extensive local progress; they differ in whether that progress closes into a unique solution.

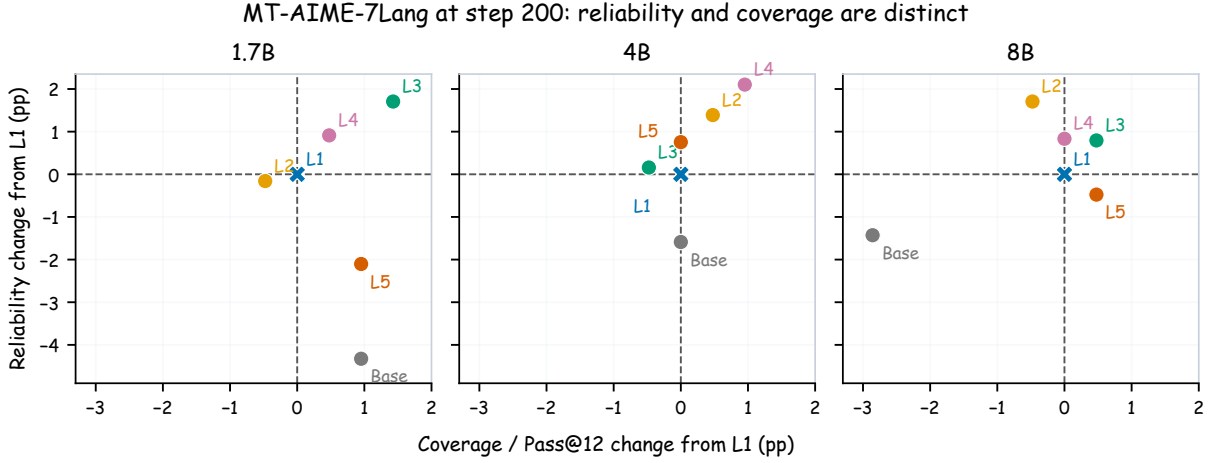


Figure 9: MT-AIME-7Lang reliability and coverage at the fixed checkpoint at step 200. Both axes report percentage-point changes from the same-scale L1 condition, so the three panels share a directly comparable coordinate system. The upper-right quadrant improves both per-generation reliability and Pass@12 coverage; points above but left of the origin improve reliability without expanding the set of problems reached by twelve samples.

Table 19: Matched AIME24 behavior at step 200. I^* is the strongest intermediate condition on the full AIME24 evaluation at step 200 (L3 for 1.7B; L4 for 4B and 8B). Avg@12 uses all 360 samples. Behavior percentages use the judge-stratified subset of 30 problems and up to four valid generations per problem and condition.

Model	Condition	Avg@12	Complete	Premature	Verify	Backtrack	Correction
1.7B	L1	53.89	65.8	37.5	69.2	47.5	33.3
	$I^* = L3$	55.00	70.0	31.7	74.2	50.0	40.0
	L5	53.61	70.0	32.5	69.2	60.0	47.5
4B	L1	73.33	79.2	19.2	83.3	36.7	28.3
	$I^* = L4$	76.11	85.0	17.5	85.8	30.8	28.3
	L5	73.89	80.8	18.3	80.8	41.7	32.5
8B	L1	78.61	85.0	18.3	87.5	35.0	30.0
	$I^* = L4$	78.33	86.7	14.2	88.3	30.0	26.7
	L5	76.67	83.2	16.0	88.2	36.1	35.3

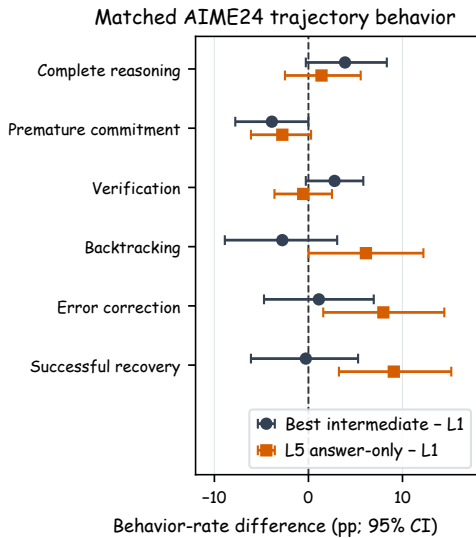


Figure 10: Matched AIME24 differences in model-judged behavior rates relative to L1 at step 200, pooled across scales, for the highest-scoring intermediate context and for answer-only L5 (95% bootstrap intervals over problems). Pooling buys interval estimates at the cost of resolution by scale; the underlying rates for each scale appear in Table 19.

Clue 6 states “there are two houses between the photography enthusiast and the person who loves beach vacations,” which in this benchmark’s convention requires positional distance three. The L3 trajectory parses the constraint correctly at first contact (“if *photography* is in house *Y*, then *beach* is in $Y+3$ or $Y-3$ ”), which collapses the search space to two placements and lets the remaining deductions close the grid. The L1 trajectory instead adopts a distance-two reading, accepts photography in house 2 with beach in house 4, and ratifies the misreading in its final self-check (“Clue 6: two houses between photography (House 2) and beach (House 4). Yes, Houses 3 between. [...] All clues are satisfied!”).

Because the misread constraint underdetermines the grid, the L1 trajectory reaches its final answer with two cells still unresolved and commits by appeal to uniqueness rather than by deduction:

```
If I assign House 1: mountain and House 3: cruise, that's one possibility. Alternatively,
House 1: cruise and House 3: mountain. Both are possible. But since the problem requires a
unique solution, and I have no more clues, I'll proceed with one of them. Let me choose House
1: mountain and House 3: cruise.
```

The trajectory verifies each clue against its own misreading, carries the misread constraint through to the end, and then guesses between alternatives it has not resolved. The failing L1 trajectory is twice as long and contains more explicit contradiction and re-derivation episodes (six “contradiction” mentions versus one). In this example, success turns on correct constraint representation and closure, not response length or visible checking activity. The selection rule and the remaining 58 contrast cases are part of the analysis artifact in the code repository.

E. Context Assignment

The 4B prompt-control runs replace every level-specific label, delimiter, and transition (Appendix B.2) with the following text:

```
Here is additional information for this problem:
=== Additional Information Begin ===
{hint}
=== Additional Information End ===

Use the additional information above to independently solve the problem.
Do not merely copy or cite it. Reason step by step, justify the necessary steps, and put the
final answer within \boxed{}
```

The student prompt is unchanged and never contains the hint. Relative to the primary 4B runs, only the teacher-side wrapper and transition are made common; the underlying L1–L5 contexts, 29,434 training problems, optimizer settings, checkpoint schedule, and evaluators remain fixed.

Table 20: Qwen3-4B shared-bridge and mixture ablations. ID columns report the best Avg@12 up to step 200 (selected step in parentheses); MT-AIME and GPQA use step 200. Mixtures assign one hint per problem.

Condition	AIME24	AIME25	HMMT25	ID mean	MT-AIME	GPQA
Base	75.00	64.44	41.39	60.28	65.87	53.69
L1: solution	75.83 (100)	68.33 (50)	44.72 (150)	62.96	66.75	<u>55.51</u>
L2: strategy	77.22 (150)	68.89 (150)	<u>47.50</u> (150)	64.54	<u>69.40</u>	55.25
L3: framing	77.22 (175)	68.61 (150)	46.11 (150)	63.98	68.57	55.45
L4: category	<u>78.06</u> (100)	<u>71.39</u> (200)	<u>46.39</u> (25)	<u>65.28</u>	68.57	55.45
L5: answer	76.11 (150)	70.00 (175)	45.56 (175)	63.89	67.34	<u>55.61</u>
Uniform L1–L4	76.11 (150)	68.89 (50)	45.56 (150)	63.52	67.74	55.15
MidMix L2/L3	76.67 (50)	<u>70.83</u> (125)	<u>46.39</u> (175)	<u>64.63</u>	<u>69.52</u>	55.20
ExtremeMix L1/L4	75.83 (125)	69.72 (100)	46.11 (200)	63.89	67.02	54.75

Each mixture contains 29,434 rows with exactly one hint per problem, deterministically under seed 20260720: Uniform uses 7,359 L1, 7,359 L2, 7,358 L3, and 7,358 L4 rows, MidMix 14,717 L2 and 14,717 L3, ExtremeMix 14,717 L1 and 14,717 L4. They test balanced exposure under a fixed example budget, not

concatenated contexts. Routed variants reuse these assignments but invoke each row’s source-level wrapper and transition, so an L1 row is formatted exactly as a primary L1 example. Each mixture is trained once at 4B and 8B under the primary configuration for that scale, 200 steps. *Routed* therefore denotes deterministic prompt construction, not a learned router or test-time selection.

Per-sample policies use Qwen3-4B, the level-specific bridges, and the training configuration of Table 4; only the context assignment varies. The routing signal is the *hint advantage* $\Delta_\ell = \log q_\ell(y) - \log q_0(y)$, the mean sampled-token log-probability advantage of a hinted teacher over a hint-free teacher baseline q_0 on the same rollout.

We selected it offline on 1,200 training problems with four base-model rollouts each. The absolute student–teacher log-probability gap separates levels weakly (0.306–0.325 nats) and teacher mass on the student’s top- k support saturates above 0.978 even at $k=4$, whereas Δ_ℓ has a median between- to within-level variance ratio of 4.5 against 1.6 for that gap, and a per-problem argmax stable across rollouts for 78% of problems. Its per-level means (−0.020, −0.027, −0.042, −0.084 for L3, L4, L2, L1) are negative throughout, lowest for L1, and match the fixed-level ordering.

Four policies assign one teacher context per trajectory: (i) *learned router*: a two-layer classifier over 39 post-rollout features, trained online every ten optimizer steps from oracle labels on the current batch; (ii) *likelihood-advantage routing*: each step scores all four hinted contexts plus q_0 (five extra no-gradient forwards) and routes by $\arg \max_\ell \Delta_\ell$; (iii) *aligned rule*: correct rollouts (by a boxed-answer check against the training answer) to L3 and incorrect ones to L4, following the offline difficulty direction; (iv) *reversed rule*: correct to L3 and incorrect to L1, on the intuition that failure needs the strongest supervision. The rules use no extra teacher forwards.

Table 21: Per-sample adaptive granularity selection at 4B (Avg@12). Per-benchmark columns and the ID mean report the best checkpoint up to step 200; ID@200 reports the mean at the fixed endpoint. Fixed L1 and L3 repeat the corresponding Table 1 conditions. No adaptive policy exceeds the best fixed level on the peak mean, and the reversed rule falls below fixed L1.

Condition	AIME24	AIME25	HMMT25	ID mean	ID@200
Fixed L1	76.94	68.61	44.72	63.43	60.93
Fixed L3(best fixed)	77.50	69.72	47.22	64.81	61.76
Learned router	76.67	71.11	46.39	64.72	62.69
Likelihood-advantage routing (Δ_ℓ)	76.94	69.44	46.11	64.17	62.87
Rule: correct→L3, incorrect→L4	76.11	70.83	46.67	64.54	62.59
Rule: correct→L3, incorrect→L1	75.56	67.78	45.83	63.06	61.76

The shared bridge preserves the separation between L1 and intermediate contexts but moves the nominal leader from L3 to L4: the family-level contrast is stable, the exact L2–L4 ranking is not.

No mixture and no per-sample policy exceeds the best fixed level on the ID peak mean (Tables 20 and 21), including likelihood-advantage routing that computes Δ_ℓ exactly at every step.

On transfer, the comparison depends on the reference: MidMix reaches 69.52 on MT-AIME, above the best fixed level with the shared bridge (L2, 69.40) but below the best fixed level with the primary bridges (L4, 69.56 in Table 9), and no mixture leads on GPQA.

Several policies improve the fixed endpoint at step 200: five of six routed mixtures exceed the corresponding best fixed-level endpoint by 0.37–0.84 points while one trails by 0.18 (Table 22; Figure 11), and the learned and likelihood-advantage routers exceed fixed L3’s endpoint by 0.93 and 1.11 points while leaving the best observed peak unchanged. Direction matters too: sending incorrect rollouts to L1 rather than L4 lowers the peak mean by 1.48 points and falls below fixed L1. These are single runs, so they suggest reduced sensitivity to the stopping point rather than a higher attainable peak.

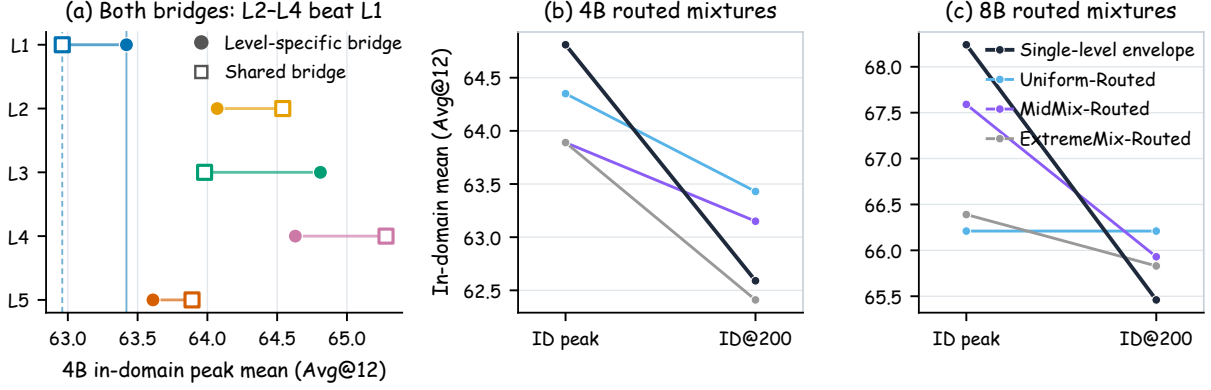


Figure 11: Controls for prompt wording and mixtures. **(a)** Shared-bridge replication: 4B in-domain peak mean (Avg@12) for each level under the level-specific and the level-agnostic bridge template. L2–L4 exceed L1 under both templates, so the separation persists under common teacher wording. **(b, c)** Routed mixtures under the level-specific bridge at 4B and 8B, compared against the single-level envelope. Five of the six routed mixtures exceed the best fixed-level endpoint at step 200, while none exceeds the best fixed-level in-domain peak.

Table 22: Level-conditioned mixture controls. ID peak averages the benchmark-wise best Avg@12 up to step 200; ID@200 averages all three ID benchmarks at the common terminal checkpoint. MT-AIME and GPQA use step 200. The single-level envelope takes the best L1–L5 result independently in each column and may therefore select different levels. Routed rows use the original level-specific bridge associated with each example’s assigned level. Light-blue/bold shading marks the column maximum within scale; light-gray/underline shading marks the second-best.

Scale	Condition	ID peak	ID@200	MT-AIME	GPQA
4B	Base	60.28	–	65.87	53.69
	Single-level envelope	64.81	62.59	69.56	55.96
	Uniform-Routed	<u>64.35</u>	63.43	68.29	54.85
	MidMix-Routed	63.89	<u>63.15</u>	67.70	<u>55.30</u>
	ExtremeMix-Routed	63.89	62.41	<u>69.13</u>	55.05
8B	Base	62.13	–	70.48	60.05
	Single-level envelope	68.24	65.46	73.61	62.68
	Uniform-Routed	66.21	66.21	72.22	<u>61.62</u>
	MidMix-Routed	<u>67.59</u>	<u>65.93</u>	<u>72.90</u>	61.01
	ExtremeMix-Routed	66.39	65.83	72.26	61.11